

A negatively pinched Kähler threefold without bounded holomorphic coordinates

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Abstract

We construct a contractible domain in complex dimension three with a complete Kähler metric whose real sectional curvatures lie between two finite negative constants. It admits no bounded holomorphic map to $\mathbb{C}^3$ with nowhere-vanishing Jacobian and is therefore not biholomorphic to a bounded domain. This gives a negative answer to the negatively pinched Kähler uniformization question.

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1 Introduction

A bounded domain in $\mathbb{C}^n$ has bounded holomorphic coordinate functions whose differentials are independent everywhere. The negatively curved uniformization problem asks whether curvature and completeness can force a complex manifold to possess such coordinates. In complex dimension one, a complete simply connected surface whose curvature is bounded above by a negative constant is biholomorphic to the disk. In higher dimension, the existence of bounded holomorphic coordinates is a separate global analytic requirement.

An early formulation appears in H. Wu's work on normal families [15, p. 195, question (1)]. He asks whether a complete simply connected Kähler manifold with nonpositive real sectional curvature and holomorphic sectional curvature bounded above by a negative constant is biholomorphic to a bounded domain. Wu and Yau state the two-sided real-sectional version with a bounded pseudoconvex domain as conclusion [13, Conjecture 4.3, p. 18]: if a simply connected complete Kähler manifold has every real sectional curvature between two negative constants, must it be biholomorphic to a bounded domain in its complex dimension?

The unit ball is already too restrictive a candidate in higher dimension. Mostow and Siu constructed compact negatively curved Kähler surfaces not covered by the ball [10]. Deraux reviews their construction and obtains three-dimensional examples [3]. Their lifted metrics on the universal covers have two-sided negative sectional bounds by compactness. These non-ball results, however, do not exclude a biholomorphism to some other bounded domain; that is the stronger obstruction sought here.

Here and throughout, sectional curvature refers to the underlying real Riemannian metric, and completeness means geodesic completeness. A bounded holomorphic map is a holomorphic map with bounded image in its Euclidean target.

Theorem 1.1. *There exist a contractible domain $M \subset \mathbb{C}^3$, a smooth complete Kähler metric g on M , and finite constants $0 < A \leq B$ such that*

$$-B \leq K_g(\sigma) \leq -A < 0$$

for every real two-plane $\sigma \subset T_p M$ and every $p \in M$. No bounded holomorphic map $F : M \rightarrow \mathbb{C}^3$ has a nowhere-vanishing Jacobian determinant. In particular, M is not biholomorphic to a bounded domain in $\mathbb{C}^3$.

Theorem 1.1 gives a negative answer to the stated uniformization question, and its curvature hypotheses also place the example within Wu's earlier question. The example has two bounded nonconstant holomorphic functions, supplied by its base coordinates. Its analytic obstruction concerns a fixed triple of bounded functions with independent differentials at every point. The domain is nevertheless Stein: the construction supplies a smooth strictly plurisubharmonic exhaustion (Lemma 7.1), so Grauert's solution of the Levi problem applies [4, Theorem 2 and Section 3.4]. Thus the obstruction is to bounded holomorphic coordinates, not to holomorphic convexity.

1.1 Intrinsic metrics and bounded functions

Two-sided negative real sectional curvature has strong consequences for intrinsic complex geometry. Greene and Wu established the lower comparison for the Bergman metric [5]; Wu and Yau recall that result and prove the complementary upper bound [14, Theorems 4 and 6]. Under completeness and simple connectivity, Wu and Yau then compare the Bergman, Kobayashi–Royden, complete negative Kähler–Einstein, and original metrics [14, Corollary 7]. The Bergman metric uses square-integrable holomorphic top-degree forms. These comparisons do not supply a fixed tuple of bounded holomorphic functions with

independent differentials everywhere. Applied to the example in Theorem 1.1, they show that all these intrinsic metrics can be complete and mutually comparable while no such tuple exists.

The existence of even one nonconstant bounded holomorphic function is a related question with a different conclusion. The companion paper [11, Theorem 1.1] constructs, in a sufficiently large fixed finite complex dimension, a domain diffeomorphic to Euclidean space with a complete Kähler metric of real sectional curvature at most -1 and only constant bounded holomorphic functions. Its curvature is unbounded below. Here both curvature bounds are finite and uniform, while the two bounded base coordinates remain. The two constructions and their proofs are independent.

1.2 Construction and proof strategy

We construct M as a family of discs over the unit ball $\mathbb{B} = \{z \in \mathbb{C}^2 : |z| < 1\}$. A smooth real function ϕ on $\mathbb{B}$ determines

$$M_\phi = \{(z, w) \in \mathbb{B} \times \mathbb{C} : e^{\phi(z)}|w|^2 < 1\}.$$

Thus the disc above z has radius $e^{-\phi(z)/2}$. Put $\psi(z) = -\log(1 - |z|^2)$. Our metric is the complex Hessian of

$$\lambda\psi(z) - \log(1 - e^{\phi(z)}|w|^2),$$

with the real-metric convention specified in Section 2. This is a specialization of Calabi's construction from a base potential and a radial function of a Hermitian fiber norm [2, Section 3, equations (3.1)–(3.2)]. The momentum formulation of Hwang and Singer develops this framework for general base and bundle data [9].

For the logarithmic radial potential used here, Hao, Chen, and Wang prove holomorphic and real sectional pinching results when the bundle curvature determines the base metric [6, Theorems 1 and 2]. Hao, Chen, Wang, and Zhang obtain related results for complete Kähler–Einstein disk-bundle metrics under Einstein and curvature hypotheses on the base [7, Sections 4.2–4.3]. The estimate needed here concerns all real two-planes when the base form $\lambda\partial\bar{\partial}\psi$ and the fiber-weight curvature $\partial\bar{\partial}\phi$ are independent. We assume upper and lower bounds for the latter form and bounds for the component derivatives needed for curvature, expressed in centered ball coordinates. No curvature sign for the metric $\partial\bar{\partial}\phi$ is assumed. We derive and estimate the full curvature tensor under precisely these hypotheses.

The analytic task is to choose ϕ so that it has these derivative bounds but cannot dominate a holomorphic real part, even after adding the logarithmic boundary allowance forced by Cauchy's estimate for a Jacobian. To do this, we construct radii $r_N \uparrow 1$ and positive probabilities μ_N on the unit sphere $S \subset \mathbb{C}^2$ such that

$$\int_S \left(\phi(r_N \xi) + 4 \log \frac{1}{1 - r_N} \right) d\mu_N(\xi) \longrightarrow -\infty,$$

while $\int_S \operatorname{Re} H(r_N \xi) d\mu_N(\xi) = \operatorname{Re} H(0)$ for every holomorphic H on $\mathbb{B}$. The two averages exclude any holomorphic real part bounded above by the tested function plus a constant. The probabilities preserve holomorphic evaluation at the origin, but such measures need not preserve the logarithmic mean inequality for holomorphic functions: Hedenmalm gives an explicit example of this distinction [8, Theorem 3.7].

The construction proceeds at degrees $k_j = Q^j$ for one fixed large integer Q . At each degree we sum homogeneous peaks centered on a separated set of complex lines. The projective packing and power-sum estimate have a direct antecedent in Ryll and Wojtaszczyk [12, Section 2]. A radial correction around each peak annihilates every holomorphic moment

exactly. Separation of the degrees makes the corrections compatible with positivity at every stage. Regularized logarithms built from the peak polynomials then have a fixed negative average, which survives all subsequent corrections. Summing these logarithms with a sufficiently large fixed coefficient makes their negative averages dominate the boundary allowance. Low degrees vary little in a centered chart, while high degrees are exponentially small; this gives the uniform derivatives through order four needed for curvature.

The geometric task has its own delicate point. To prove negative curvature, we must show that its negated numerator is positive on every real two-plane. After rescaling the base directions, the leading expression is nonnegative but can vanish on planes with nonzero base and fiber projections. On these limiting planes the contribution of the potentially largest error term vanishes as well. This cancellation gives stronger error control near the degenerate planes and proves strict negativity for one sufficiently large base parameter λ . With λ fixed, compactness on bounded fiber ranges and a uniform limiting tensor at the fiber boundary give both finite curvature bounds.

Section 2 states the two precise ingredients and derives Theorem 1.1 from them by Cauchy estimates for a putative Jacobian. Sections 3–4 construct the peaks and positive representing densities; Sections 5–6 establish their logarithmic tests and the controlled potential. Section 7 proves completeness. Sections 8–9 compute the curvature and prove the two-sided estimate, including the limiting planes just described.

2 Two inputs and the bounded-map obstruction

The construction separates a problem about functions on the ball from a curvature estimate. The first input is a strictly plurisubharmonic function with controlled derivatives and an obstruction to a prescribed holomorphic real-part bound. The second turns its complex Hessian bounds into a complete metric with uniformly negative real sectional curvature. We state both inputs precisely and then prove their implication for Theorem 1.1.

2.1 Centered charts and the two propositions

Let $\mathbb{B} = \{z \in \mathbb{C}^2 : |z| < 1\}$ and put

$$\psi(z) = -\log(1 - |z|^2), \quad I = (\psi_{i\bar{j}}).$$

Here a complex Hessian is also viewed as a Hermitian form, linear in its first argument. To center coordinates at a point $z_0 \in \mathbb{B}$, let $r = |z_0|$, choose a unitary map U with $U(r, 0) = z_0$, and set

$$T = U \circ T_r, \quad T_r(\zeta) = \left(\frac{r + \zeta_1}{1 + r\zeta_1}, \frac{\sqrt{1 - r^2}\zeta_2}{1 + r\zeta_1} \right). \quad (2.1)$$

These maps are ball automorphisms with $T(0) = z_0$. The identity

$$1 - |T_r(\zeta)|^2 = \frac{(1 - r^2)(1 - |\zeta|^2)}{|1 + r\zeta_1|^2}$$

shows that T^*I has the same expression in ζ as I has in z : the additional terms in $\psi \circ T$ are constant or pluriharmonic. In particular, this matrix is the identity at 0 and has zero first derivatives there. We call (2.1) the centered ball charts.

For a smooth real function ϕ on $\mathbb{B}$, write $h = \partial\bar{\partial}\phi$ and, in a centered chart T , write

$$h_{i\bar{j}}^T(\zeta) = \partial_{\zeta_i} \partial_{\bar{\zeta}_j} (\phi \circ T)(\zeta).$$

The uniform control needed below is the existence of constants $0 < c \leq C < \infty$ such that, for every centered chart,

$$c \text{Id} \leq h^T(0) \leq C \text{Id} \quad (2.2)$$

and

$$\left| \partial_k h_{ij}^T(0) \right| + \left| \partial_{\bar{\ell}} h_{ij}^T(0) \right| + \left| \partial_k \partial_{\bar{\ell}} h_{ij}^T(0) \right| \leq C \quad (i, j, k, \ell \in \{1, 2\}). \quad (2.3)$$

The derivatives here are ordinary derivatives of the displayed component functions. In invariant form, (2.2) says $cI \leq h \leq CI$.

Proposition 2.1 (A controlled potential). *There exist $\phi \in C^\infty(\mathbb{B}, \mathbb{R})$ and a finite constant C for which (2.2) and (2.3) hold with $c = 1$, and such that no $H \in \mathcal{O}(\mathbb{B})$ and $C_H \in \mathbb{R}$ satisfy*

$$\text{Re } H(z) \leq C_H + 4 \log \frac{1}{1 - |z|} + \phi(z) \quad (z \in \mathbb{B}). \quad (2.4)$$

For any smooth real ϕ on $\mathbb{B}$, define its Hartogs domain by

$$M_\phi = \{(z, w) \in \mathbb{B} \times \mathbb{C} : t(z, w) < 1\}, \quad t(z, w) = e^{\phi(z)} |w|^2. \quad (2.5)$$

For $\lambda > 0$ define the potential and its complex Hessian by

$$\mathcal{K}_\lambda = \lambda \psi - \log(1 - t), \quad \hat{g}_\lambda = \partial \bar{\partial} \mathcal{K}_\lambda. \quad (2.6)$$

Our real metric convention is $g_\lambda = 2 \text{Re } \hat{g}_\lambda$: if a real vector X has $(1, 0)$ part u , then $g_\lambda(X, X) = 2 \hat{g}_\lambda(u, u)$.

Proposition 2.2 (Hartogs metric transfer). *Let $\phi \in C^\infty(\mathbb{B}, \mathbb{R})$ satisfy (2.2) and (2.3) for fixed $0 < c \leq C < \infty$. There is a finite $\lambda_0 = \lambda_0(c, C) > 0$ such that, for every $\lambda \geq \lambda_0$, the domain M_ϕ in (2.5) is contractible and the tensor $g_\lambda = 2 \text{Re } \hat{g}_\lambda$ defined by (2.6) is a smooth geodesically complete Kähler metric. For this fixed λ there are constants $0 < A_\lambda \leq B_\lambda < \infty$ such that*

$$-B_\lambda \leq K_{g_\lambda}(\sigma) \leq -A_\lambda < 0$$

for every point of M_ϕ and every real two-plane σ in its tangent space.

The proof of Proposition 2.1 occupies the analytic construction below. The global geometry and curvature calculation in Sections 7–9 prove Proposition 2.2. We first establish the short argument that connects their conclusions to bounded holomorphic maps.

2.2 The Jacobian on the zero section

Lemma 2.3. *Let $\phi \in C^\infty(\mathbb{B}, \mathbb{R})$ have the nonexistence property in Proposition 2.1: no holomorphic H and real constant C_H satisfy (2.4). Then no bounded holomorphic map $F : M_\phi \rightarrow \mathbb{C}^3$ has a nowhere-vanishing complex Jacobian determinant.*

Proof. Suppose that $F = (F_1, F_2, F_3)$ is such a map, and choose $B \geq 1$ so that $|F_\ell| \leq B$ on M_ϕ for every component. Since M_ϕ is open in $\mathbb{C}^3$, its coordinates (z_1, z_2, w) are global. The function

$$d(z) = \det \frac{\partial(F_1, F_2, F_3)}{\partial(z_1, z_2, w)}(z, 0)$$

is holomorphic and nowhere zero on $\mathbb{B}$.

Fix $z \in \mathbb{B}$ and put $\delta = 1 - |z|$. The restriction $F_\ell(\cdot, 0)$ is bounded and holomorphic on the entire base ball. For $j = 1, 2$, the disk $\{z + \zeta e_j : |\zeta| < \delta\}$ is contained in that ball. Cauchy's estimate, applied on disks with radii tending to δ , gives

$$|\partial_{z_j} F_\ell(z, 0)| \leq B \delta^{-1}. \quad (2.7)$$

For this fixed z , the fiber in M_ϕ is the disk of radius $e^{-\phi(z)/2}$. Cauchy's estimate on this disk gives

$$|\partial_w F_\ell(z, 0)| \leq B e^{\phi(z)/2}. \quad (2.8)$$

In particular, the base estimate requires no uniform fiber radius near z ; it is taken entirely on the zero section.

The six terms in the determinant expansion, together with (2.7) and (2.8), yield

$$|d(z)| \leq 6B^3 \delta^{-2} e^{\phi(z)/2},$$

and hence

$$\log |d(z)|^2 \leq 2 \log 6 + 6 \log B + 4 \log \frac{1}{1-|z|} + \phi(z). \quad (2.9)$$

Because $\mathbb{B}$ is simply connected and d has no zeros, it has a global holomorphic logarithm ℓ . For example, a primitive of the closed holomorphic one-form $d^{-1} \partial d$, with a suitable constant, satisfies $e^\ell = d$. The holomorphic function $H = 2\ell$ has $\operatorname{Re} H = \log |d|^2$, so (2.9) contradicts (2.4). $\square$

Proof of Theorem 1.1. Choose ϕ from Proposition 2.1, and then choose λ as in Proposition 2.2. The domain $M = M_\phi \subset \mathbb{C}^3$ is contractible and carries the asserted smooth complete Kähler metric with two uniform negative real-sectional bounds. Lemma 2.3 excludes a bounded holomorphic map to $\mathbb{C}^3$ with nowhere-vanishing Jacobian. A biholomorphism to a bounded domain would be such a map: the chain rule applied to its holomorphic inverse makes its Jacobian invertible everywhere. $\square$

The logarithm argument uses nonvanishing along the entire zero section. It gives no assertion that a bounded map's determinant vanishes identically. Nor are all bounded holomorphic functions constant: z_1 and z_2 are nonconstant bounded holomorphic functions on M_ϕ .

3 Separated homogeneous peaks

We begin the construction of the potential in Proposition 2.1. Its analytic requirement will follow from increasingly negative averages against positive measures that preserve holomorphic means. The following observation explains the role of those measures before we construct them.

Write $S = \{\xi \in \mathbb{C}^2 : |\xi| = 1\}$, and let σ be its surface measure normalized to have mass one. A probability measure μ on S represents holomorphic evaluation at zero if

$$\int_S p(\xi) d\mu(\xi) = p(0) \quad \text{for every holomorphic polynomial } p.$$

Lemma 3.1 (Representing-mean criterion). *Let u be a continuous real function on $\mathbb{B}$. Suppose there are radii $0 < r_N < 1$ and representing probabilities μ_N on S such that*

$$\int_S u(r_N \xi) d\mu_N(\xi) \longrightarrow -\infty.$$

Then no holomorphic function H on $\mathbb{B}$ and constant C_H satisfy $\operatorname{Re} H \leq C_H + u$ on $\mathbb{B}$.

Proof. For each fixed $r_N < 1$, the Taylor polynomials of H converge uniformly on $r_N S$. The representing identities therefore give

$$\int_S \operatorname{Re} H(r_N \xi) d\mu_N(\xi) = \operatorname{Re} H(0).$$

Integrating the proposed inequality contradicts the assumed limit. $\square$

We will apply this criterion to $u(z) = \phi(z) + 4 \log(1/(1 - |z|))$. The first ingredient is a supply of homogeneous polynomials with many separated peaks. Their uniform bounds will also control the derivatives of the eventual potential.

3.1 Projective packing and polynomial bounds

The projective distance and separated-power estimate used here occur in the direct construction of Ryll and Wojtaszczyk [12, p. 112 and Lemma 2.7]. We give the packing argument with a Gaussian tail and an adjustable spacing parameter, since the density corrections will require both. Related constructions with peaks localized in nonisotropic sphere balls appear in Aleksandrov [1, Lemmas 2–3]; the caps used here are invariant under a common phase rotation.

Identify complex lines in $\mathbb{C}^2$ with points of $\mathbb{CP}^1$. For unit representatives define

$$d([\xi], [p]) = \sqrt{1 - |\langle \xi, p \rangle|^2},$$

where the Hermitian inner product is linear in its first argument. This equals the operator norm distance between the rank-one orthogonal projections onto the two lines: their difference has trace zero and eigenvalues $\pm \sqrt{1 - |\langle \xi, p \rangle|^2}$. The triangle inequality for the operator norm therefore proves that d is a metric. We also write $d(\xi, p)$ when the representatives are understood.

In unitary coordinates with $\xi_1 = \langle \xi, p \rangle$, write

$$\xi = (e^{i\theta} \cos v, e^{i\gamma} \sin v), \quad 0 \leq v \leq \pi/2.$$

The round volume element is $\cos v \sin v \, dv \, d\theta \, d\gamma$. After normalization and the change of variable $u = \cos^2 v$, this gives

$$d\sigma = du \frac{d\theta}{2\pi} \frac{d\gamma}{2\pi}, \quad 0 \leq u \leq 1. \quad (3.1)$$

In particular, a projective cap $d(\xi, p) < s$ has measure s^2 for $0 \leq s \leq 1$.

Fix $D_0 \geq 1$. For an integer $k \geq D_0^2$, choose a maximal $D_0/\sqrt{k}$ -separated set of lines, and choose one unit representative p for each line. Let $\mathcal{P}_k$ denote this set of representatives and $N_k = |\mathcal{P}_k|$. Define the homogeneous holomorphic polynomial

$$P_k(z) = \sum_{p \in \mathcal{P}_k} \langle z, p \rangle^k. \quad (3.2)$$

The representatives may be chosen arbitrarily; the estimates below hold for every such choice.

Lemma 3.2 (Peak bounds). *There are absolute constants $C, c > 0$, independent of $D_0 \geq 1$, such that*

$$\frac{k}{D_0^2} \leq N_k \leq \frac{4k}{D_0^2} \quad (3.3)$$

and, for every $\xi \in S$ and $u \geq 0$,

$$\sum_{\substack{p \in \mathcal{P}_k \\ \sqrt{k}d(\xi, p) \geq u}} |\langle \xi, p \rangle|^k \leq C e^{-cu^2}. \quad (3.4)$$

Consequently,

$$|P_k(z)| \leq C |z|^k \quad (z \in \mathbb{C}^2), \quad \int_S |P_k| \, d\sigma \leq \frac{2N_k}{k+2}. \quad (3.5)$$

Proof. Put $\rho = D_0/\sqrt{k} \leq 1$. Maximality gives a covering by caps of radius ρ , and separation gives disjoint interiors for caps of radius $\rho/2$. Their measures yield $1 \leq N_k \rho^2$ and $N_k \rho^2/4 \leq 1$, proving (3.3).

We also need a local count whose constant does not depend on D_0 . About the centers within distance $u/\sqrt{k}$ of a fixed line, place caps of radius $1/(3\sqrt{k})$. Their interiors are disjoint, and they lie inside the cap of radius $(u + 1/3)/\sqrt{k}$ about that line. Comparing measures bounds their number by $9(u + 1/3)^2$ when the latter radius is at most one. Otherwise their number is at most $9k \leq 9(u + 1/3)^2$, by comparison with the whole space. Thus the local count is at most $C(1 + u)^2$ in all cases.

Since

$$|\langle \xi, p \rangle|^k = (1 - d(\xi, p)^2)^{k/2} \leq e^{-kd(\xi, p)^2/2},$$

the contribution from the annulus $n \leq \sqrt{k}d(\xi, p) < n + 1$ is at most $C(n + 2)^2 e^{-n^2/2}$. Summing these bounds from $n = \lfloor u \rfloor$ onward proves (3.4): the polynomial factor is absorbed by a weaker Gaussian exponent, and the range $0 \leq u \leq 2$ is absorbed by the constant. Setting $u = 0$ and using homogeneity gives the first assertion in (3.5), including points outside the unit ball. Finally, (3.1) gives

$$\int_S |\langle \xi, p \rangle|^k d\sigma = \int_0^1 u^{k/2} du = \frac{2}{k+2}.$$

The triangle inequality proves the stated integral bound. $\square$

3.2 Patches and a radial profile

For a fixed $R_0 > 0$ and a center $p \in \mathcal{P}_k$, use the adapted coordinates above and put

$$y = -k \log |\xi_1|, \quad \mathcal{U}_{k,p} = \{\xi \in S : 0 \leq y < R_0\}. \quad (3.6)$$

The set $y = 0$ is the phase circle through p . Every patch is invariant under multiplication of ξ by a common phase.

Lemma 3.3 (Patch estimates). *Fix $R_0 > 0$. If $D_0 > \max\{1, 2\sqrt{2R_0}\}$ and $k \geq \max\{D_0^2, 2R_0\}$, the patches $\mathcal{U}_{k,p}$ are pairwise disjoint. On each patch,*

$$d\sigma = \frac{2}{k} e^{-2y/k} dy \frac{d\theta}{2\pi} \frac{d\gamma}{2\pi}, \quad (3.7)$$

and

$$|P_k(\xi) - e^{-y} e^{ik\theta}| \leq C e^{-c(D_0 - \sqrt{2R_0})^2}. \quad (3.8)$$

Outside their union,

$$|P_k(\xi)| \leq C e^{-cR_0}. \quad (3.9)$$

The constants C, c can be chosen independently of R_0, D_0, k .

Proof. The projective radius of a patch is $\sqrt{1 - e^{-2R_0/k}} \leq \sqrt{2R_0/k}$, so the separation of the centers makes the patches disjoint. Formula (3.7) follows from $|\xi_1|^2 = e^{-2y/k}$ in (3.1).

If $\xi \in \mathcal{U}_{k,p}$ and $p' \neq p$ is another center, the metric triangle inequality gives

$$\sqrt{k}d(\xi, p') \geq D_0 - \sqrt{2R_0}.$$

The peak from p is exactly $e^{-y} e^{ik\theta}$. Summing all other peaks by (3.4) proves (3.8). Outside every patch, each center satisfies

$$kd(\xi, p)^2 \geq k(1 - e^{-2R_0/k}) \geq R_0.$$

The final inequality uses $k \geq 2R_0$ and $1 - e^{-v} \geq v/2$ for $0 \leq v \leq 1$. Another application of (3.4) proves (3.9). $\square$

The density correction will use a radial profile with an exact weighted cancellation. The following choice also supplies the strict inequality needed for the logarithmic tests later.

Lemma 3.4 (Radial profiles). *Fix $1 < a < 2$ and put $\alpha = \log a$. One can choose $0 < s_1 < 1$, smooth real functions f, b on $[0, \infty)$, and $R_0 > 0$ such that*

$$\alpha^2 - 2s_1(\cosh \alpha - 1) < 0, \quad (3.10)$$

$$\|f\|_\infty < 1, \quad f = -s_1 \text{ on } [0, \alpha], \quad \int_0^\infty e^{-y} f(y) dy = 0, \quad b' - b = f, \quad b(0) = 0. \quad (3.11)$$

Both functions vanish on a neighborhood of $[R_0, \infty)$. Moreover, R_0 can be chosen so that $|aP_k| \leq 1/2$ off the patches whenever the hypotheses of Lemma 3.3 hold.

Proof. The strict inequality $2(\cosh \alpha - 1) > \alpha^2$ allows a choice

$$\frac{\alpha^2}{2(\cosh \alpha - 1)} < s_1 < 1.$$

Since $a < 2$, this choice also satisfies $s_1(1 - e^{-\alpha}) < e^{-\alpha}$. Choose a small $\delta > 0$ and a smooth cutoff χ , equal to one on $[0, \alpha]$, between zero and one, and zero for $y \geq \alpha + \delta$. For sufficiently small δ ,

$$N = s_1 \int_0^\infty e^{-y} \chi(y) dy < e^{-(\alpha+2\delta)}.$$

Choose a smooth compactly supported function ν , supported strictly after $\alpha + 2\delta$, with $0 \leq \nu \leq 1$, and with

$$P = \int_0^\infty e^{-y} \nu(y) dy > N.$$

Such a function is obtained by approximating the indicator of a sufficiently long interval beginning just after $\alpha + 2\delta$. Then $f = -s_1\chi + (N/P)\nu$ has the required plateau and weighted integral. The two supports are disjoint, so $\|f\|_\infty \leq \max\{s_1, N/P\} < 1$.

Define

$$b(y) = e^y \int_0^y e^{-v} f(v) dv.$$

It satisfies $b' - b = f$ and $b(0) = 0$. Beyond the support of f , the integral is exactly zero by the weighted cancellation, so b also has compact support. Near zero, $b(y) = -s_1(e^y - 1)$, which is smooth up to the boundary of the half-line. Smoothness here does not require extending f by zero to negative arguments. Finally choose R_0 larger than both supports, and enlarge it further until $aCe^{-cR_0} \leq 1/2$ in (3.9). $\square$

The plateau $0 \leq y \leq \alpha$ is exactly the region where the model peak ae^{-y} has modulus at least one. For the profiles just constructed, the integral of $\log |1 - ae^{-y}e^{it'}|^2$ against $(1 + f(y) \cos t') dy dt' / (2\pi)$ on $[0, R_0] \times [0, 2\pi]$ equals $\alpha^2 - 2s_1(\cosh \alpha - 1)$, as calculated in (5.7). Thus the strict profile inequality prescribes a negative logarithmic integral, while the weighted cancellation will preserve holomorphic moments.

Fix these profiles and R_0 from now on. The spacing D_0 and the degrees will remain available to be chosen sufficiently large. In particular, enlarging D_0 later improves the single-peak approximation without altering the profiles.

4 Positive representing densities

We now construct the representing probabilities required by Lemma 3.1. Each step adds an oscillation on the current patches. A radial integration by parts makes every holomorphic moment of that oscillation vanish exactly. The separation between consecutive degrees will ensure that the new density stays positive.

Keep the profiles f, b, R_0 from Lemma 3.4 fixed, and put

$$c_0 = \frac{1 - \|f\|_\infty}{4} > 0.$$

Fix any $D_0 > \max\{1, 2\sqrt{2R_0}\}$. For an integer $Q \geq 2$ large enough that every degree below meets the patch thresholds, put

$$k_j = Q^j, \quad \ell_0 = 0, \quad \ell_j = \sum_{l=1}^j k_l \quad (j \geq 1). \quad (4.1)$$

Thus $\ell_{j-1} < k_j$ and $\ell_j < 2k_j$. At each degree use any of the separated sets and patches constructed in Section 3.

Start with $W_0 = 1$. At stage j , set $k = k_j$. For each $p \in \mathcal{P}_k$ let

$$S_p(\theta) = W_{j-1}(e^{i\theta}p).$$

To determine the correction, consider a phase mode $e^{is\theta}$ with $|s| \leq \ell_{j-1}$, the frequency range maintained below. Multiplication by $e^{ik\theta}$ shifts it to frequency $m = k + s > 0$; its conjugate can pair with the holomorphic monomial $\xi_1^m = e^{-my/k}e^{im\theta}$. After canceling the surface-measure factor $e^{-2y/k}$, its radial moment will vanish if its coefficient is $b' - (m/k)b$, because

$$e^{-my/k} \left(b' - \frac{m}{k}b \right) = \frac{d}{dy} \left(e^{-my/k}b \right), \quad b(0) = b(R_0) = 0.$$

The identity $f = b' - b$ therefore prescribes the coefficient $f - (s/k)b$ for this mode. Summing the modes leads to the following definition in the adapted coordinates of $\mathcal{U}_{k,p}$:

$$A_{j,p}(\xi) = e^{2y/k} \operatorname{Re} \left[e^{ik\theta} \left(f(y)S_p(\theta) - b(y) \frac{S'_p(\theta)}{ik} \right) \right], \quad (4.2)$$

$$W_j = W_{j-1} + \sum_{p \in \mathcal{P}_k} A_{j,p},$$

where each correction is extended by zero off its patch. The prime on S_p denotes differentiation in θ . This recursion is defined without assuming positivity; that property will be proved for one fixed sufficiently large Q .

Lemma 4.1 (Smoothness of the corrections). *Every W_j defined by (4.2) is a smooth real function on S .*

Proof. Proceed by induction, beginning with $W_0 = 1$. The restriction S_p is a smooth periodic function. To check the central circle of a patch, use the smooth coordinates

$$(\theta, \zeta) \mapsto (e^{i\theta} \sqrt{1 - |\zeta|^2}, \zeta), \quad y = -\frac{k}{2} \log(1 - |\zeta|^2).$$

All terms of (4.2) are smooth in these coordinates, including at $\zeta = 0$. They do not depend on the transverse polar angle γ , which would be singular there. Their dependence on θ is periodic, so it defines a smooth function along the full central circle. At the outer boundary, f and b vanish in a collar of R_0 ; hence the correction is identically zero in a collar of the patch boundary. Its extension by zero is therefore smooth. Summing the finitely many corrections proves the induction. $\square$

Let T denote the phase derivative on S ,

$$TW(\xi) = \frac{d}{d\omega} W(e^{i\omega}\xi) \Big|_{\omega=0}.$$

A horizontal vector $Z \in T_\xi S$ will always mean a *Euclidean unit* real tangent vector satisfying the complex equation $\langle Z, \xi \rangle = 0$.

A function has phase bandwidth at most ℓ if, on every orbit $\omega \mapsto e^{i\omega}\xi$, it is a trigonometric polynomial with frequencies in $[-\ell, \ell]$. Its orbitwise mean is the constant coefficient of this polynomial.

Theorem 4.2 (Uniform representing densities). *There is a constant $K_* > 0$, depending only on the fixed profiles f, b and R_0 , with the following property. For every fixed $D_0 > \max\{1, 2\sqrt{2R_0}\}$ there is an integer threshold Q_0 such that, for every integer $Q \geq Q_0$ and every choice of the separated sets above, the recursion (4.2) has the following properties for all $j \geq 1$:*

1. W_j is smooth and strictly positive, and $W_j d\sigma$ represents holomorphic evaluation at zero. In particular it is a probability measure.
2. Under global phase rotation, W_j has frequencies only between $-\ell_j$ and ℓ_j , and its mean on every phase orbit is exactly one.
3. At every point of S ,

$$\begin{aligned} c_0 W_{j-1} &\leq W_j \leq 2W_{j-1}, \\ |ZW_j| &\leq K_* \sqrt{k_j} W_j, \quad |TW_j| \leq K_* k_j W_j \end{aligned} \tag{4.3}$$

for every unit horizontal vector Z .

The threshold Q_0 may also be enlarged to impose any fixed lower threshold on the degrees k_j .

We first prove the exact moment and frequency assertions; they hold before positivity is known. The remainder of the theorem, including the order of choices of K_* and Q_0 , is proved in the next subsection.

4.1 Exact moments and phase frequencies

Lemma 4.3 (Moment preservation). *For every stage of (4.2), the phase bandwidth of W_j is at most ℓ_j , its orbitwise mean is one, and*

$$\int_S p(\xi) W_j(\xi) d\sigma(\xi) = p(0) \quad \text{for every holomorphic polynomial } p. \tag{4.4}$$

Each individual correction has zero integral against every such polynomial, including the constant polynomial.

Proof. At stage zero the phase assertions are immediate. The polynomial identity for σ follows by global phase averaging: every nonconstant homogeneous holomorphic polynomial has nonzero positive phase frequency.

Assume the assertions through stage $j-1$, and write $k = k_j$. The central restriction has a finite expansion

$$S_p(\theta) = \sum_{|s| \leq \ell_{j-1}} \widehat{S}_p(s) e^{is\theta}, \quad \widehat{S}_p(0) = 1.$$

Since $b' - b = f$, the bracket in (4.2) is

$$\sum_s \widehat{S}_p(s) \left(b' - \frac{k+s}{k} b \right) e^{is\theta}.$$

All integers $m = k + s$ are positive because $\ell_{j-1} < k$. The patch and y are invariant under global phase rotation, whereas θ increases by the rotation angle. Hence the correction has frequencies in

$$[k - \ell_{j-1}, k + \ell_{j-1}] \cup [-k - \ell_{j-1}, -k + \ell_{j-1}]. \quad (4.5)$$

They omit zero. Multiplying by a function of y and extending by zero across the invariant patch boundary do not alter these frequencies. The bandwidth of W_j is thus at most $k + \ell_{j-1} = \ell_j$, and its orbitwise mean remains one.

For the full moment calculation, use an adapted monomial $\xi_1^n \xi_2^l$, with nonnegative integers n, l . If $l > 0$, averaging in the transverse phase γ gives zero. If $l = n = 0$, averaging in θ gives zero because all $m = k + s$ are positive. Suppose then that $l = 0$ and $n \geq 1$. As b is real, the correction is

$$A_{j,p} = \frac{e^{2y/k}}{2} \sum_s \left(\widehat{S}_p(s) e^{i(k+s)\theta} + \overline{\widehat{S}_p(s)} e^{-i(k+s)\theta} \right) \left(b' - \frac{k+s}{k} b \right).$$

Since $\xi_1^n = e^{-ny/k} e^{in\theta}$, only the conjugate frequency with $k + s = n$ can survive the θ integration. Formula (3.7) gives exactly

$$\begin{aligned} \int_{\mathcal{U}_{k,p}} \xi_1^n A_{j,p} d\sigma &= \frac{\overline{\widehat{S}_p(n-k)}}{k} \int_0^{R_0} e^{-ny/k} \left(b' - \frac{n}{k} b \right) dy \\ &= \frac{\overline{\widehat{S}_p(n-k)}}{k} \left[e^{-ny/k} b(y) \right]_0^{R_0} = 0. \end{aligned}$$

Here a coefficient outside the previous bandwidth is zero. The factor $2/k$ in surface measure combines with the $1/2$ from the real part to leave $1/k$. Both endpoint terms vanish because $b(0) = b(R_0) = 0$. The argument applies to every degree n , without a degree cutoff. Unitary changes of coordinates preserve holomorphic polynomials, so each correction annihilates every polynomial in the original coordinates. Adding the corrections proves (4.4) and completes the induction. $\square$

In particular, every W_j has total integral one. What remains for Theorem 4.2 is to prove that these exact representing densities are positive with the uniform derivative bounds. The next estimates accomplish this with a single choice of Q , independent of the number of stages.

4.2 Uniform positivity and derivative estimates

We now prove the remaining assertions of Theorem 4.2. The main point is to choose one derivative constant K_* , followed by one threshold for Q , that works at every stage. A phase derivative of the correction in (4.2) involves S_p'' . We obtain a bound for this derivative from the frequency support and the estimates already proved at earlier stages.

Lemma 4.4 (Growth along phase orbits). *Suppose $Q \geq 2$, $k_j = Q^j$, and the positive functions $W_0 = 1, W_1, \dots, W_n$, with $n \geq 1$, satisfy*

$$c_0 W_{j-1} \leq W_j \leq 2W_{j-1}, \quad |TW_j| \leq K_* k_j W_j \quad (1 \leq j \leq n).$$

Put

$$R = \frac{2}{c_0}, \quad \beta = \log_2 R.$$

For every $p \in S$, the function $S_p(\theta) = W_n(e^{i\theta}p)$ satisfies

$$S_p(\theta + u) \leq e^{K_*} R(1 + k_n|u|)^\beta S_p(\theta), \quad |u| \leq \pi. \quad (4.6)$$

In particular, the constants in this bound are independent of Q, n, p .

Proof. Choose the largest $1 \leq l \leq n$ such that $k_l|u| \leq 1$, or put $l = 0$ if no such index exists. The comparisons between successive densities give

$$\frac{W_n(e^{i(\theta+u)}p)}{W_n(e^{i\theta}p)} \leq R^{n-l} \frac{W_l(e^{i(\theta+u)}p)}{W_l(e^{i\theta}p)}.$$

When $l \geq 1$, logarithmic differentiation and the phase derivative bound show that the last ratio is at most $\exp(K_* k_l|u|) \leq e^{K_*}$. For $l = 0$ it equals one. If $l = n$, this already proves the assertion. Otherwise the maximality of l , including the case $l = 0$, gives

$$k_n|u| > Q^{n-l-1} \geq 2^{n-l-1}.$$

It follows that $R^{n-l} \leq R(1 + k_n|u|)^\beta$, proving (4.6). $\square$

Lemma 4.5 (A weighted derivative bound). *Fix $A \geq 1$ and $\beta \geq 0$. Let $q \geq 1$ be an integer, and let s be a positive trigonometric polynomial of degree at most $2q$. Suppose that*

$$s(\theta + u) \leq A(1 + q|u|)^\beta s(\theta) \quad (\theta \in \mathbb{R}, |u| \leq \pi).$$

There is a constant $C(A, \beta)$, independent of q, s , such that

$$|s''(\theta)| \leq C(A, \beta) q^2 s(\theta).$$

Proof. Choose a fixed smooth compactly supported function $\chi : \mathbb{R} \rightarrow \mathbb{R}$ which equals one on $[-2, 2]$, and put

$$K_q(u) = \sum_{h \in \mathbb{Z}} \chi(h/q) e^{ihu}.$$

For normalized convolution on the circle, the frequency assumption implies $s'' = s * K_q''$. We claim that, for every positive integer N ,

$$|K_q''(u)| \leq C_N q^3 (1 + q|u|)^{-N}, \quad |u| \leq \pi, \quad (4.7)$$

where C_N depends only on N, χ .

Indeed the coefficients of K_q'' are

$$a_h = -h^2 \chi(h/q) = q^2 g(h/q), \quad g(v) = -v^2 \chi(v).$$

There are $O(q)$ nonzero coefficients, each of size $O(q^2)$, so direct summation gives $|K_q''| \leq Cq^3$. An N -th finite difference of a_h is an iterated integral of the N -th derivative of $q^2 g(v/q)$, and hence has size at most $C_N q^{2-N}$. Only $O_N(q)$ such differences can be nonzero, since g is compactly supported and $q \geq 1$. Their absolute sum is therefore at most $C_N q^{3-N}$. Multiplication of the Fourier series by $(1 - e^{iu})^N$ produces these differences, up to an index shift. Using $|1 - e^{iu}| \geq 2|u|/\pi$ on $[-\pi, \pi]$ gives the second estimate

$$|K_q''(u)| \leq C_N q^{3-N} |u|^{-N} \quad (u \neq 0).$$

Combining it with the direct estimate proves (4.7).

Choose an integer $N > \beta + 1$. Positivity, the assumed growth bound, and (4.7) now give

$$\begin{aligned} |s''(\theta)| &\leq \int_{-\pi}^{\pi} s(\theta - u) |K_q''(u)| \frac{du}{2\pi} \\ &\leq AC_N q^3 s(\theta) \int_{-\pi}^{\pi} (1 + q|u|)^{\beta-N} \frac{du}{2\pi} \\ &\leq C(A, \beta) q^2 s(\theta). \end{aligned}$$

□

Combining these two lemmas with the bandwidth assertion of Lemma 4.3 gives the following consequence. If the estimates of Theorem 4.2 hold through stage $j - 1$, then for $k = k_j$ and the central function S_p in (4.2),

$$\frac{|S_p'|}{kS_p} \leq \frac{K_*}{Q}, \quad \frac{|S_p''|}{k^2 S_p} \leq \frac{C_1}{Q^2}, \quad (4.8)$$

where C_1 depends only on K_*, c_0 and the fixed cutoff χ . It is independent of Q, j, p . For $j \geq 2$, apply the lemmas with $q = k_{j-1}$, noting that $\ell_{j-1} \leq 2k_{j-1}$; the first inequality follows directly from the phase derivative bound. For $j = 1$, both derivatives vanish because $S_p = 1$.

We next estimate a single correction using only normalized bounds. This will allow us to choose K_* before requiring (4.8) to be small. All horizontal vectors Z below have Euclidean length one. Enlarge a constant $D_R \geq 1$, depending only on R_0 , so that on every patch at every sufficiently large degree k ,

$$|Yy| \leq D_R, \quad |Y\theta| \leq D_R/k, \quad |Y(e^{ik\theta})| \leq D_R, \quad Y = Z/\sqrt{k} \text{ or } Y = T/k. \quad (4.9)$$

Here is a direct verification. With $c = \langle \xi, p \rangle$, complex horizontality gives

$$|Zc| \leq \sqrt{1 - |c|^2} \leq \sqrt{2R_0/k}, \quad |c| \geq e^{-R_0/k}.$$

Differentiate $y = -k \log |c|$ and $\theta = \arg c$ to obtain the first two bounds for $Z/\sqrt{k}$; the third follows from the second. For T/k , one has $Yy = 0$ and $Y\theta = 1/k$. The fixed lower threshold for k makes $e^{-R_0/k} \geq 1/2$. We also choose D_R large enough to bound the length of the radial arc from $e^{i\theta}p$ to each patch point by $D_R/\sqrt{k}$. In adapted coordinates this arc is

$$u \mapsto (e^{i\theta} \cos u, e^{i\gamma} \sin u), \quad 0 \leq u \leq \arccos(e^{-y/k}).$$

It has unit speed and is complex-horizontal. The length bound follows from $0 \leq y \leq R_0$ and $1 - e^{-y/k} \leq R_0/k$. Fix this enlarged D_R throughout the rest of the proof.

Lemma 4.6 (Normalized correction estimate). *Write*

$$F = \|f\|_{\infty}, \quad B = \|b\|_{\infty}, \quad F_1 = \|f'\|_{\infty}, \quad B_1 = \|b'\|_{\infty}.$$

On a degree- k patch suppose $W_{j-1} > 0$, $S_p > 0$, and

$$\frac{e^{2y/k} S_p}{W_{j-1}} \leq 2, \quad r_1 = \frac{|S_p'|}{kS_p} \leq 1, \quad r_2 = \frac{|S_p''|}{k^2 S_p} \leq 1.$$

For the correction $A_{j,p}$ in (4.2),

$$|YA_{j,p}| \leq C_2 W_{j-1}, \quad Y = Z/\sqrt{k} \text{ or } T/k, \quad C_2 = 2D_R[4(F + B) + F_1 + B_1].$$

In particular, C_2 is independent of K_, Q, j, p .*

Proof. Besides (4.9), use

$$|Y(e^{2y/k})| \leq (2D_R/k)e^{2y/k}, \quad |YS_p| \leq (D_R/k)|S'_p|, \quad |YS'_p| \leq (D_R/k)|S''_p|.$$

Differentiating (4.2) and taking absolute values gives

$$|YA_{j,p}| \leq D_R e^{2y/k} S_p [(1 + 2/k)(F + Br_1) + F_1 + B_1 r_1 + Fr_1 + Br_2].$$

Since $k \geq 1$, the assumed normalized bounds imply the stated estimate. $\square$

Completion of the proof of Theorem 4.2. Fix $K_* > 2C_2/c_0$. The constant C_1 in (4.8) is now fixed as well. We give a finite list of requirements on Q , all independent of the induction stage.

The horizontal derivative estimate at stage $j - 1$, integrated along the radial arc just described, will imply

$$e^{-K_* D_R / \sqrt{Q}} \leq \frac{S_p(\theta)}{W_{j-1}(\xi)} \leq e^{K_* D_R / \sqrt{Q}}. \quad (4.10)$$

Indeed the logarithmic derivative along a unit horizontal arc is at most $K_* \sqrt{k_{j-1}}$, and the arc length is at most $D_R / \sqrt{k_j}$. At the initial stage the ratio is exactly one. Since $k_j \geq Q$, put

$$H(Q) = \exp \left(K_* D_R / \sqrt{Q} + 2R_0 / Q \right).$$

Choose a threshold Q_0 such that every integer $Q \geq Q_0$ satisfies all earlier degree and patch requirements, and also

$$Q \geq 2, \quad H(Q) \leq 2, \quad K_*/Q \leq 1, \quad C_1/Q^2 \leq 1, \quad Q^{-1/2} \leq c_0/2, \quad (4.11)$$

$$H(Q)(F + BK_*/Q) \leq 1 - 2c_0.$$

These conditions are compatible: $H(Q) \rightarrow 1$ and $F = 1 - 4c_0 < 1 - 2c_0$.

We now fix any integer $Q \geq Q_0$ and induct on j . Assume the claimed estimates through stage $j - 1$; at $j = 1$ the preceding density is the constant one. Equation (4.8), obtained from earlier stages only, and the horizontal comparison (4.10) establish the normalized hypotheses of Lemma 4.6. Moreover,

$$|A_{j,p}| \leq e^{2y/k} S_p(F + Br_1) \leq H(Q)(F + BK_*/Q)W_{j-1} \leq (1 - 2c_0)W_{j-1}.$$

At any point at most one patch correction is nonzero. Hence

$$2c_0 W_{j-1} \leq W_j \leq (2 - 2c_0)W_{j-1},$$

which proves positivity and the required multiplicative comparison.

For $Y = Z/\sqrt{k_j}$, the previous derivative bound gives $|YW_{j-1}| \leq (K_*/\sqrt{Q})W_{j-1}$. For $Y = T/k_j$, it gives the smaller coefficient K_*/Q . Together with Lemma 4.6, (4.11), and the choice of K_* , these yield

$$|YW_j| \leq (C_2 + K_*/\sqrt{Q})W_{j-1} \leq K_* c_0 W_{j-1} \leq K_* W_j.$$

Outside the patches the correction vanishes. Its smooth zero extension at their boundaries gives the same estimate there. Multiplying by $\sqrt{k_j}$ or k_j proves the two derivative bounds and completes the induction.

Lemma 4.3 supplies the exact moments, rotation mean and bandwidth independently of these inequalities. In particular $\int_S W_j d\sigma = 1$; positivity therefore makes $W_j d\sigma$ a probability measure. $\square$

We emphasize the order of the choices. The normalized correction constant C_2 depends only on the fixed patch data. It determines K_* ; this in turn determines C_1 ; finally Q_0 is chosen. No lower bound for W_j uniform in j is required. The positive comparison at each stage suffices for logarithmic differentiation, and all later integral estimates will use the probability normalization.

5 Regularized logarithmic tests

Keep the profile parameters $a, \alpha, s_1, f, b, R_0$ from Lemma 3.4. We shall construct a smooth plurisubharmonic function at each degree whose average against its representing density is uniformly negative. The same negative bound will hold against all later densities. The estimates are uniform for all sufficiently large integers Q , so the final value of Q can be chosen after the regularization and spacing parameters.

A positive measure representing holomorphic evaluation need not satisfy the logarithmic inequality $\log |f(0)| \leq \int_S \log |f| d\mu$ for every holomorphic f continuous on the closed ball. Hedenmalm gives an explicit example of this distinction between representing and Jensen measures [8, Theorem 3.7]. Here we need smooth logarithmic tests and negative bounds that persist through all later density corrections. We prove these properties directly for the densities of Theorem 4.2.

Fix a constant C_P in the bound $|P_k(z)| \leq C_P |z|^k$ of Lemma 3.2.

For $\epsilon > 0$, define

$$F_\epsilon(v) = \log \frac{|1 - av|^2 + \epsilon^2}{1 + \epsilon^2}, \quad U_k(z) = F_\epsilon(P_k(z)). \quad (5.1)$$

In particular $F_\epsilon(0) = U_k(0) = 0$. The function U_k is smooth and plurisubharmonic on $\mathbb{C}^2$, since

$$(U_k)_{i\bar{j}} = \frac{a^2 \epsilon^2}{(|1 - aP_k|^2 + \epsilon^2)^2} (P_k)_i \overline{(P_k)_j}. \quad (5.2)$$

For fixed ϵ and $A < \infty$, all real derivatives of F_ϵ of any fixed order are bounded on $|v| \leq A$. The mean-value estimate also gives

$$|F_\epsilon(v)| \leq C_{\epsilon,A} |v| \quad (|v| \leq A). \quad (5.3)$$

All subsequent regularization constants may depend on the fixed ϵ .

Lemma 5.1 (Exact averaging). *Let $k = k_j = Q^j$, where Q satisfies the hypotheses of Theorem 4.2. For $0 \leq r \leq 1$,*

$$\int_S U_k(r\xi) W_j(\xi) d\sigma = \int_S U_k(r\xi) d\sigma + \sum_p \int_{\mathcal{U}_{k,p}} U_k(r\xi) e^{2y/k} f(y) \cos(k\theta) d\sigma. \quad (5.4)$$

Proof. Under the global phase rotation $\xi \mapsto e^{i\omega} \xi$, homogeneity gives $P_k(e^{i\omega} \xi) = e^{ik\omega} P_k(\xi)$. Thus $U_k(r\xi)$ has only phase frequencies in $k\mathbb{Z}$. By Lemma 4.3, the phase bandwidth of W_{j-1} is $\ell_{j-1} < k$ and its constant coefficient on each orbit is one. Averaging the product over the phase therefore gives

$$\int_S U_k(r\xi) W_{j-1}(\xi) d\sigma = \int_S U_k(r\xi) d\sigma.$$

Each patch is phase-invariant. Its radial variable y is fixed under rotation, while θ and γ both increase by ω . Write the central restriction as $S_p(\theta) = W_{j-1}(e^{i\theta} p) = \sum_s \hat{S}_p(s) e^{is\theta}$.

In the patch increment of (4.2), the frequencies are $\pm(k+s)$ with $|s| \leq \ell_{j-1} < k$. Pairing with a frequency in $k\mathbb{Z}$ is possible only for $s = 0$. Since $\hat{S}_p(0) = 1$ and S'_p has zero constant coefficient, the surviving part of the increment is exactly $e^{2y/k} f(y) \cos(k\theta)$. Integrating this phase average proves (5.4). $\square$

The next estimate chooses the regularization before the spacing. This order matters: the single-peak approximation will be applied to a fixed smooth logarithm, and its error can then be made small by increasing D_0 .

Proposition 5.2 (Negative averages at one scale). *There exist $\epsilon > 0$, $\eta \in (0, 1/2)$ and $D_* \geq 1$, depending only on the fixed profile data, with the following property. For every $D_0 \geq D_*$ there are $k_* > 0$ and $d_2 > 0$ such that, for every integer Q satisfying the density theorem and $Q \geq k_*$,*

$$\int_S U_{k_j}(r\xi) W_j(\xi) d\sigma \leq -d_2 \quad (j \geq 1, \quad 1 - \eta \leq r^{k_j} \leq 1). \quad (5.5)$$

The constants k_*, d_2 may depend on D_0 , but are independent of j and of further increases in Q .

Proof. Put

$$\Delta = 2s_1(\cosh \alpha - 1) - \alpha^2 > 0.$$

We first compute the integral for a single peak:

$$I_\epsilon(\rho) = \int_0^{R_0} \int_0^{2\pi} \log \frac{|1 - a\rho e^{-y} e^{it'}|^2 + \epsilon^2}{1 + \epsilon^2} (1 + f(y) \cos t') \frac{dt'}{2\pi} dy. \quad (5.6)$$

At $\rho = 1$, the unregularized logarithm has only one singularity, at $(y, t') = (\alpha, 0)$ modulo 2π . Near that point,

$$|1 - e^{\alpha - y + it'}|^2 \asymp (y - \alpha)^2 + (t')^2.$$

Its logarithm is locally integrable. For $0 < \epsilon \leq 1$, the regularized logarithm is bounded above uniformly, and its negative part is bounded by the negative part of the unregularized logarithm plus $\log 2$. Dominated convergence therefore applies in (5.6).

For $q < 1$, the mean and cosine moment of $\log |1 - qe^{it'}|^2$ are 0 and $-q$, respectively, by the logarithm series. For $q > 1$, factoring out q^2 gives mean $2 \log q$ and cosine moment $-q^{-1}$. Using the cancellation $\int_0^{R_0} e^{-y} f(y) dy = 0$, we obtain

$$\begin{aligned} I_0(1) &= \alpha^2 - \int_0^\alpha f(y) e^{y-\alpha} dy - \int_\alpha^{R_0} f(y) e^{\alpha-y} dy \\ &= \alpha^2 + \int_0^\alpha f(y) (e^{\alpha-y} - e^{-(\alpha-y)}) dy \\ &= \alpha^2 - 2s_1(\cosh \alpha - 1) = -\Delta. \end{aligned} \quad (5.7)$$

The last equality uses $f = -s_1$ on $[0, \alpha]$.

We now compare this model with (5.4). For $k = k_j$ put $\rho = r^k$, so homogeneity gives $U_k(r\xi) = F_\epsilon(\rho P_k(\xi))$. Off the patches, $|aP_k| \leq 1/2$ by Lemma 3.4. If $v = \rho P_k(\xi)$, $0 \leq \rho \leq 1$, and $s = |1 - av|^2$, then $1/4 \leq s \leq 9/4$ and

$$F_\epsilon(v) - \log |1 - av|^2 = \log(1 + \epsilon^2/s) - \log(1 + \epsilon^2).$$

The derivative in s of the first term has magnitude at most $16\epsilon^2$, and $|s - 1| \leq C|v|$. Hence

$$|F_\epsilon(v) - \log |1 - av|^2| \leq C\epsilon^2 |P_k(\xi)|.$$

The off-patch set is phase-invariant, and the phase mean of the unregularized logarithm is zero. By the L^1 estimate of Lemma 3.2, its total contribution is thus at most

$$C_{\text{off}} \epsilon^2 \frac{N_k}{k}. \quad (5.8)$$

Here C_{off} is independent of D_0, k, ρ and $0 < \epsilon \leq 1$.

On a patch, the measure formula of Lemma 3.3 writes the combined contribution as

$$\frac{2}{k} \int_0^{R_0} \int_{\mathbb{T}^2} F_\epsilon(\rho P_k(\xi)) (e^{-2y/k} + f(y) \cos(k\theta)) \frac{d\theta d\gamma}{(2\pi)^2} dy.$$

For fixed ϵ, R_0 , replace $P_k(\xi)$ by $e^{-y} e^{ik\theta}$ using the single-peak estimate, and replace $e^{-2y/k}$ by one. The bounded derivative of F_ϵ and the bound $|e^{-2y/k} - 1| \leq 2R_0/k$ show that the result differs from $(2/k)I_\epsilon(\rho)$ by at most

$$\frac{2}{k} C_{\epsilon, R_0} \left(e^{-c(D_0 - \sqrt{2R_0})^2} + \frac{1}{k} \right). \quad (5.9)$$

All constants are uniform over patches and $0 \leq \rho \leq 1$. The change from $k\theta$ to t' in the angular average is exact, as k is an integer.

Choose $\epsilon \in (0, 1]$ so small that

$$|I_\epsilon(1) + \Delta| \leq \Delta/8, \quad C_{\text{off}} \epsilon^2 \leq \Delta/8.$$

By continuity of (5.6) in ρ , choose $\eta \in (0, 1/2)$ such that $I_\epsilon(\rho) \leq -3\Delta/4$ for $1 - \eta \leq \rho \leq 1$. Next choose D_* large enough for patch disjointness and for the first error in (5.9) to be at most $\Delta/16$ inside its factor $2/k$. For each $D_0 \geq D_*$, take k_* large enough for all the geometric estimates and for the remaining error inside this factor to be at most $\Delta/16$. Summing the patches and adding (5.8) gives, in particular,

$$\int_S U_k(r\xi) W_j d\sigma \leq -\frac{\Delta}{2} \frac{N_k}{k} \leq -\frac{\Delta}{2D_0^2}.$$

Thus $d_2 = \Delta/(2D_0^2)$ is a valid choice. $\square$

Lemma 5.3 (Negative averages against later densities). *Fix $\epsilon, \eta, D_0, k_*, d_2$ as in Proposition 5.2, and put $d_3 = d_2/2$. There is a threshold Q_* such that, for every integer $Q \geq Q_*$, the densities satisfy*

$$\int_S U_{k_j}(r\xi) W_N(\xi) d\sigma \leq -d_3 \quad (N \geq j \geq 1, \quad 1 - \eta \leq r^{k_j} \leq 1). \quad (5.10)$$

Proof. Include the density and degree thresholds in Q_* , and require $Q \geq 4$. By Lemma 4.3, the increment $W_\nu - W_{\nu-1}$ has frequencies within $\ell_{\nu-1}$ of $\pm k_\nu$. Since $\ell_{\nu-1} \leq 2k_{\nu-1}$, each such frequency has absolute value at least $(Q-2)k_{\nu-1}$. Summing the increments with $\nu > j$ shows that every frequency in $W_N - W_j$ has absolute value at least $(Q-2)k_j$.

For fixed ξ, r , write the logarithmic profile as

$$V(t') = \log \frac{|1 - Ae^{it'}|^2 + \epsilon^2}{1 + \epsilon^2}, \quad A = ar^{k_j} P_{k_j}(\xi).$$

The uniform polynomial bound gives $|A| \leq aC_P$. Differentiating the logarithm shows that

$$\|V''\|_\infty \leq B_\epsilon := \frac{2aC_P}{\epsilon^2} + \frac{4a^2C_P^2}{\epsilon^4}.$$

Two integrations by parts bound its m th Fourier coefficient by B_ϵ/m^2 for $m \neq 0$. To track these coefficients as functions of ξ , define

$$c_m(\xi) = \int_0^{2\pi} F_\epsilon(r^{k_j} P_{k_j}(\xi) e^{it'}) e^{-imt'} \frac{dt'}{2\pi}.$$

Homogeneity and a change of variable give $c_m(e^{i\omega}\xi) = e^{imk_j\omega} c_m(\xi)$. Thus c_m has global phase frequency mk_j , and $U_{k_j}(r\xi) = \sum_{m \in \mathbb{Z}} c_m(\xi)$. The coefficient bound makes this series uniformly convergent. Its truncation to $|m| \leq Q-3$ has uniform error at most $2B_\epsilon/(Q-3)$ and only frequencies of absolute value less than $(Q-2)k_j$. Phase averaging therefore makes its integral against $W_N - W_j$ vanish. Both densities are positive probabilities by Theorem 4.2; hence $\|W_N - W_j\|_{L^1(\sigma)} \leq 2$. It follows that

$$\left| \int_S U_{k_j}(r\xi) (W_N - W_j) d\sigma \right| \leq \frac{4B_\epsilon}{Q-3}.$$

Choose Q_* so that this is at most $d_2/2$, and apply (5.5). $\square$

6 The controlled base potential

We now sum the logarithmic tests to prove Proposition 2.1. Fix parameters and an integer Q for which Theorem 4.2 and Lemma 5.3 hold, and write $k_j = Q^j$. For a finite amplitude $L > 0$, set

$$\phi(z) = \psi(z) + L \sum_{j=1}^{\infty} U_{k_j}(z). \quad (6.1)$$

We first justify this definition. We then choose L to exclude the required pluriharmonic minorant, and finally prove the uniform chart estimates needed by the metric construction.

Lemma 6.1 (Smooth convergence). *For every finite $L > 0$, the series in (6.1) converges locally in $C^\infty(\mathbb{B})$. Its sum is smooth, satisfies $\phi(0) = 0$, and has complex Hessian $h \geq I$.*

Proof. Fix $0 < r_0 < r_1 < 1$. Around each point of $|z| \leq r_0$, the complex polydisk of radius $(r_1 - r_0)/(2\sqrt{2})$ lies in $|z| < r_1$. Cauchy estimates there, together with Lemma 3.2, give

$$\sup_{|z| \leq r_0} |D^m P_k(z)| \leq C_{m,r_0,r_1} r_1^k$$

for every fixed derivative order m . Here and below D^m denotes any real partial derivative of total order m . The values of P_k remain in a fixed bounded disk. Every positive-order derivative of $F_\epsilon(P_k)$ is a sum of products containing at least one positive-order derivative of P_k or its conjugate. The bounded derivatives of F_ϵ therefore give the same exponential estimate for $D^m U_k$. For $m = 0$, use (5.3). Since $\sum_j r_1^{Q^j} < \infty$, the series converges in every local C^m norm.

Each summand vanishes at zero and is plurisubharmonic by (5.2). Termwise differentiation is justified by the local C^2 convergence, and hence

$$\partial\bar{\partial}\phi = I + L \sum_j \partial\bar{\partial}U_{k_j} \geq I.$$

$\square$

Lemma 6.2 (Absence of a pluriharmonic minorant). *If*

$$L > \frac{5 \log Q}{d_3}, \quad (6.2)$$

there are no $H \in \mathcal{O}(\mathbb{B})$ and $C_H \in \mathbb{R}$ satisfying (2.4).

Proof. Put

$$r_N = 1 - \frac{\eta}{k_N}. \quad (6.3)$$

For every $j \leq N$, Bernoulli's inequality gives

$$r_N^{k_j} = \left(1 - \frac{\eta}{k_N}\right)^{k_j} \geq 1 - \eta \frac{k_j}{k_N} \geq 1 - \eta.$$

Thus all of the first N modes satisfy the hypothesis of (5.10) at this same radius. For $j = N + m > N$, on the other hand,

$$r_N^{k_j} \leq e^{-\eta k_j / k_N} = e^{-\eta Q^m}.$$

By the uniform polynomial bound and (5.3),

$$\sum_{j > N} \sup_{\xi \in S} |U_{k_j}(r_N \xi)| \leq C_\epsilon \sum_{m \geq 1} e^{-\eta Q^m} =: C_{\text{tail}} < \infty, \quad (6.4)$$

independently of N . Because W_N is a positive probability, the same bound controls the weighted tail, without requiring a uniform pointwise bound on W_N .

The base potential contributes only one logarithm:

$$\psi(r_N \xi) = -\log(1 - r_N^2) \leq \log \frac{1}{1 - r_N} = \log(k_N / \eta).$$

Combining this with (5.10) and (6.4) yields

$$\begin{aligned} \int_S \left(\phi(r_N \xi) + 4 \log \frac{1}{1 - r_N} \right) W_N d\sigma \\ \leq 5 \log(k_N / \eta) + L(-d_3 N + C_{\text{tail}}) \\ = N(5 \log Q - L d_3) + 5 \log(1 / \eta) + L C_{\text{tail}} \longrightarrow -\infty. \end{aligned} \quad (6.5)$$

By Theorem 4.2, the measures $\mu_N = W_N d\sigma$ are positive representing probabilities. Apply Lemma 3.1 to these measures, the radii r_N , and the continuous function $u(z) = \phi(z) + 4 \log(1 / (1 - |z|))$. The limit (6.5) is exactly its hypothesis, so the required minorant cannot exist. $\square$

It remains to control the metric jets as the chart centers approach the boundary. The value of ϕ need not be bounded. For its derivatives, the low degrees become almost constant in a centered chart, while the high degrees are exponentially small.

Lemma 6.3 (Uniform chart estimates). *For every fixed finite $L > 0$, the potential (6.1) has uniformly bounded positive-order derivatives through order four at the origins of all centered ball charts. Consequently its complex Hessian satisfies (2.2) and (2.3), with $c = 1$ and a finite constant C .*

Proof. After a unitary change of coordinates, put the center at $(r, 0)$, where $0 \leq r < 1$, and let $\delta = 1 - r$. Rotating P_k preserves its degree and its uniform homogeneous bound. Use the chart T_r of (2.1), on the fixed complex polydisk

$$D_\rho = \{|\zeta_1| < \rho, |\zeta_2| < \rho\}, \quad \rho = \frac{1}{16}.$$

The denominators in T_r have modulus at least $1 - \rho$. We estimate the polynomial in the two ranges of $k\delta$.

High degrees: $k\delta > 1$. The ball automorphism identity gives, on D_ρ ,

$$1 - |T_r(\zeta)|^2 = \frac{(1 - r^2)(1 - |\zeta|^2)}{|1 + r\zeta_1|^2} \geq \frac{1 - 2\rho^2}{(1 + \rho)^2} \delta.$$

It follows that $1 - |T_r(\zeta)| \geq c\delta$, with a fixed $c > 0$. Thus

$$|P_k(T_r(\zeta))| \leq C_P e^{-ck\delta}.$$

Cauchy estimates in this fixed polydisk bound every fixed-order holomorphic derivative at zero, including order zero, by $C_m e^{-ck\delta}$.

Low degrees: $k\delta \leq 1$. Consider the polydisk in the original coordinates

$$A_k = \left\{ |z_1 - r| < \frac{1}{k}, \quad |z_2| < \frac{1}{\sqrt{k}} \right\}.$$

It may extend outside the ball, but P_k is an entire homogeneous polynomial. The bound of Lemma 3.2 therefore holds there as well. In fact

$$|z|^2 \leq (1 + 1/k)^2 + 1/k \leq 1 + 4/k, \quad |P_k(z)| \leq C_P (1 + 2/k)^k \leq C_P e^2.$$

Cauchy estimates on the concentric half-polydisk give

$$|\partial_1 P_k| \leq Ck, \quad |\partial_2 P_k| \leq C\sqrt{k}.$$

Meanwhile the chart displacements satisfy

$$|T_r^1(\zeta) - r| \leq \frac{2\rho}{1 - \rho} \delta, \quad |T_r^2(\zeta)| \leq \frac{\sqrt{2}\rho}{1 - \rho} \sqrt{\delta}.$$

Both numerical coefficients are less than $1/2$. As $\delta \leq 1/k$, the chart image lies in the half-polydisk just used. Integrating the gradient along the straight segment from $(r, 0)$ to $T_r(\zeta)$ gives

$$|P_k(T_r(\zeta)) - P_k(r, 0)| \leq C(k\delta + \sqrt{k\delta}) \leq 2C\sqrt{k\delta}.$$

This segment stays in the half-polydisk by convexity. Apply Cauchy estimates on D_ρ to the holomorphic difference $P_k \circ T_r - P_k(r, 0)$. Every positive-order derivative at zero is bounded by $C_m \sqrt{k\delta}$. We make no such assertion for the zeroth-order value in this range.

Composition and summation. In both ranges the values of $P_k \circ T_r$ stay in a fixed bounded disk. Each positive-order real derivative of $U_k \circ T_r = F_\epsilon(P_k \circ T_r)$ is a sum of products containing differentiated factors of $P_k \circ T_r$ or its conjugate. For $1 \leq m \leq 4$, the preceding bounds and the smooth composition estimate give

$$|D^m(U_k \circ T_r)(0)| \leq C_{\epsilon, m} \begin{cases} \sqrt{k\delta}, & k\delta \leq 1, \\ e^{-ck\delta}, & k\delta > 1. \end{cases} \quad (6.6)$$

Indeed every term contains at least one differentiated factor, and a product of several factors obeys the same bound because the displayed scale factor is at most one. This proves the estimate for all real derivatives, hence for all pure and mixed complex derivatives of these orders.

Let J be the largest index with $Q^J \delta \leq 1$, or put $J = 0$ if there is no such index. The low-degree contribution is bounded by the geometric sum

$$\sum_{j \leq J} \sqrt{Q^j \delta} \leq \frac{1}{1 - Q^{-1/2}}.$$

For the remaining degrees, the first value of $Q^j \delta$ exceeds one, so

$$\sum_{j > J} e^{-cQ^j \delta} \leq \sum_{m \geq 0} e^{-cQ^m} < \infty.$$

Both bounds are uniform in the chart center. Termwise differentiation is legitimate by Lemma 6.1. Multiplication by the fixed amplitude L preserves these bounds.

Finally, the base potential transforms as

$$\psi \circ T_r = -\log(1 - |\zeta|^2) - \log(1 - r^2) + \log |1 + r\zeta_1|^2.$$

Its positive-order derivatives through order four at zero are uniformly bounded. The last term is pluriharmonic, so the complex Hessian of $\psi \circ T_r$ is exactly the standard ball Hessian in these coordinates. The derivatives of $\phi \circ T_r$ needed for the pulled-back Hessian components and their first and second derivatives have orders two, three and four. The bounds just proved, together with $h \geq I$ from Lemma 6.1, establish the assertion. $\square$

Proof of Proposition 2.1. Choose $a, \alpha, s_1, f, b, R_0$ as in Lemma 3.4. Proposition 5.2 first chooses ϵ, η and then permits any sufficiently large spacing D_0 . Fix such a D_0 and its degree threshold k_* , and fix $d_3 = d_2/2 > 0$. The density theorem holds for every Q above a threshold depending only on the fixed patch data and spacing. Its derivative constant K_* is fixed before that threshold is chosen. We may therefore choose one integer Q satisfying the density threshold, $Q \geq k_*$, and the Fourier-error threshold of Lemma 5.3.

Now fix a finite amplitude L satisfying (6.2). Lemma 6.1 constructs the smooth potential with $h \geq I$; Lemma 6.2 proves the required obstruction; and Lemma 6.3 supplies its finite uniform chart bounds. The chart constant is allowed to depend on these fixed choices, including L . The metric parameter λ in Proposition 2.2 is chosen only afterward from these bounds, so none of the choices above depends on λ . $\square$

7 The global Hartogs metric

We now turn to Proposition 2.2. In this section we prove all its global geometric assertions. The real sectional-curvature bounds will follow from the calculation in Section 8 and the estimates in Section 9.

The potential specializes Calabi's construction on a Hermitian line bundle [2, Section 3]. We verify its positivity, smoothness, and completeness directly, including at the zero section.

Lemma 7.1. *Let ϕ be a smooth real plurisubharmonic function on $\mathbb{B}$, and let $\lambda > 0$. The domain M_ϕ in (2.5) is contractible, and the complex Hessian of $\mathcal{K}_\lambda = \lambda\psi - \log(1 - e^\phi|w|^2)$ defines a smooth geodesically complete Kähler metric $g = 2\operatorname{Re} \hat{g}$ on M_ϕ . Moreover, $\mathcal{K}_\lambda$ is a smooth strictly plurisubharmonic exhaustion of M_ϕ .*

Proof. The domain and a smooth positive metric. The function $t = e^\phi|w|^2$ is smooth on $\mathbb{B} \times \mathbb{C}$, so $M_\phi = \{t < 1\}$ is open in $\mathbb{C}^3$. The map

$$(z, w) \longmapsto (z, e^{\phi(z)/2} w)$$

is a smooth real diffeomorphism from M_ϕ to $\mathbb{B} \times \mathbb{D}$, where $\mathbb{D} = \{u \in \mathbb{C} : |u| < 1\}$. Its inverse is $(z, u) \mapsto (z, e^{-\phi(z)/2}u)$. Thus M_ϕ is contractible; no holomorphic trivialization is needed for this conclusion.

The potential $\mathcal{K}_\lambda$ is smooth at every point of M_ϕ , including its zero section. Write

$$h = \partial\bar{\partial}\phi, \quad x = \frac{t}{1-t}, \quad G = \lambda I + xh, \quad \vartheta = dw + w\partial\phi.$$

The base covectors together with ϑ form a global smooth complex coframe. Differentiating the potential gives

$$\hat{g} = G + \frac{e^\phi}{(1-t)^2} \vartheta \otimes \bar{\vartheta}. \quad (7.1)$$

For a complex tangent vector $u = (u_z, u_w)$ this means

$$\hat{g}(u, u) = G(u_z, u_z) + \frac{e^\phi}{(1-t)^2} |u_w + w\partial\phi(u_z)|^2.$$

Since $h \geq 0$, the first term controls every nonzero horizontal projection through $G \geq \lambda I$; when $u_z = 0$, the second term is positive unless $u_w = 0$. Thus $\hat{g}$ is positive definite. At the zero section its matrix is particularly simple:

$$\hat{g}|_{(z,0)} = \lambda I(z) + e^{\phi(z)} dw \otimes d\bar{w}. \quad (7.2)$$

The mixed blocks vanish. Being a positive complex Hessian of a smooth real potential, $\hat{g}$ defines the stated Kähler metric.

Away from the zero section one can write $s = \log t$ and $q = x(1+x) = t/(1-t)^2$. Then $\partial s = \vartheta/w$, so (7.1) also reads

$$\hat{g} = \lambda I + xh + q\partial s \otimes \bar{\partial} s. \quad (7.3)$$

We will use this expression for the curvature computation. Formula (7.1), rather than the coordinate s , defines the metric across $w = 0$.

Two functions with bounded gradient. For a real smooth function f , the convention $g = 2 \operatorname{Re} \hat{g}$ gives

$$|\nabla f|_g^2 = 2|\partial f|_{\hat{g}}^2. \quad (7.4)$$

Indeed, for a real vector X with $(1,0)$ part u , $g(X, X) = 2\hat{g}(u, u)$ and $df(X) = 2 \operatorname{Re} \partial f(u)$, and the identity follows by taking dual norms.

Globally on M_ϕ ,

$$\partial[-\log(1-t)] = \frac{e^\phi \bar{w}}{1-t} \vartheta.$$

In the coframe of (7.1), the Hermitian metric is block diagonal. Therefore

$$|\partial[-\log(1-t)]|_{\hat{g}}^2 = \frac{|e^\phi \bar{w}/(1-t)|^2}{e^\phi/(1-t)^2} = t.$$

This calculation includes $w = 0$, where both sides are zero. For the base potential, direct inversion of its complex Hessian gives

$$|\partial\psi|_{I^{-1}}^2 = |z|^2.$$

The covector $\partial\psi$ has only base components in the same coframe, and $G \geq \lambda I$. Combining these facts with (7.4) yields

$$|\nabla[-\log(1-t)]|_g^2 = 2t < 2, \quad |\nabla\psi|_g^2 \leq \frac{2|z|^2}{\lambda} \leq \frac{2}{\lambda}. \quad (7.5)$$

Compact containment of metric balls. Fix $p \in M_\phi$ and a finite radius $R \geq 0$. Integrating (7.5) along arbitrary piecewise smooth paths and taking the infimum of their lengths shows that $d_g(p, q) \leq R$ implies

$$-\log(1 - t(q)) \leq a_R := -\log(1 - t(p)) + \sqrt{2}R, \quad \psi(q) \leq b_R := \psi(p) + \sqrt{2/\lambda}R.$$

This argument does not require a minimizing geodesic. Set

$$\tau_R = 1 - e^{-a_R} < 1, \quad \rho_R = \sqrt{1 - e^{-b_R}} < 1.$$

Every such $q = (z, w)$ satisfies $t(q) \leq \tau_R$ and $|z| \leq \rho_R$. Let m_R be the minimum of ϕ on the compact base ball $\{|z| \leq \rho_R\}$. Then

$$|w|^2 = e^{-\phi(z)}t(q) \leq e^{-m_R}\tau_R.$$

Consequently the metric ball is contained in

$$K_R = \{(z, w) : |z| \leq \rho_R, e^{\phi(z)}|w|^2 \leq \tau_R\}. \quad (7.6)$$

This set is closed and bounded in $\mathbb{C}^3$, hence compact, and the strict bounds $\rho_R < 1$ and $\tau_R < 1$ place it entirely inside M_ϕ . In particular, a bounded-length curve cannot escape through the base boundary, through a fiber boundary, or through unbounded fiber coordinates. The zero section is an interior locus with the smooth positive metric (7.2).

The same compact-containment argument applies to sublevel sets of $\mathcal{K}_\lambda$. Both ψ and $-\log(1 - t)$ are nonnegative, so $\mathcal{K}_\lambda \leq R$ implies $\psi \leq R/\lambda$ and $-\log(1 - t) \leq R$. These bounds place the sublevel set in a compact set of the form (7.6); the sublevel set is closed there, hence compact. Positivity of its complex Hessian was proved above, so $\mathcal{K}_\lambda$ is a strictly plurisubharmonic exhaustion.

Geodesic completeness. Suppose a unit-speed geodesic has a finite maximal forward time T . Its image lies in the compact set K_T . On this set the smooth positive metric has a uniform positive Euclidean lower bound, so the geodesic's Euclidean velocity is bounded. The Christoffel symbols are also bounded on K_T , and the geodesic equation then bounds its Euclidean acceleration. Both position and velocity have limits as time tends to T , with limiting position in $K_T \subset M_\phi$. Local existence for the smooth geodesic equation extends the geodesic beyond T , a contradiction. The same argument in reverse time proves geodesic completeness. $\square$

The hypotheses of Proposition 2.2 imply $h \geq cI > 0$, so Lemma 7.1 applies for every $\lambda > 0$ under consideration. It remains to choose λ large enough for the two uniform real-sectional bounds. We do this in the next two sections, keeping the convention $g = 2 \operatorname{Re} \hat{g}$ throughout.

8 Curvature on real two-planes

We retain the base potential and Hartogs metric of Proposition 2.2. Section 9 will prove the two uniform sectional-curvature bounds. Here we compute the full curvature tensor and reduce its value on an arbitrary real two-plane to a scalar expression. In particular, the calculation includes real planes that are not complex lines.

Throughout this section, constants in tensor estimates depend only on the uniform centered-chart bounds for $h = (\phi_{i\bar{j}})$ in Proposition 2.2; the base dimension is two. We take $\lambda \geq 1$. The base form remains $I = (\psi_{i\bar{j}})$, with $\psi = -\log(1 - |z|^2)$.

8.1 A pointwise holomorphic gauge and the tensor blocks

Fix a point with $w \neq 0$, choose centered ball coordinates at its base point, and choose a local branch of $\log w$. Put

$$s = \log t = \phi + \log w + \overline{\log w}, \quad x = \frac{t}{1-t}, \quad q = x(1+x).$$

In these coordinates the centered base point is zero. Remove the first and pure second derivatives of the weight by the holomorphic fiber change

$$P(z) = \sum_i \phi_i(0) z_i + \frac{1}{2} \sum_{i,k} \phi_{ik}(0) z_i z_k, \quad v = \log w + P(z).$$

More precisely, $s = v + \bar{v} + \tilde{\phi}$, where $\tilde{\phi} = \phi - P - \bar{P}$. At the point under consideration,

$$s_i = s_{ik} = 0, \quad s_v = s_{\bar{v}} = 1, \quad s_{i\bar{j}} = h_{i\bar{j}}. \quad (8.1)$$

All higher derivatives of s involving a vertical index vanish. Subtracting $P + \bar{P}$ changes neither h nor its component derivatives in the centered base coordinates.

The metric at the point consequently has blocks

$$\hat{g} = \begin{pmatrix} G & 0 \\ 0 & q \end{pmatrix}, \quad G = \lambda I + xh.$$

At the center, I is the identity matrix. Normalize the vertical vector to $q^{-1/2}\partial_v$ and denote its index by 0. Horizontal indices i, j, k, l range over 1, 2. In this pointwise frame the metric is $G + V$, where V is the Hermitian form whose only nonzero entry is $V_{0\bar{0}} = 1$. Horizontal forms are extended by zero on the vertical direction. For any Hermitian form P , define

$$(S_P)_{a\bar{b}c\bar{d}} = P_{a\bar{b}}P_{c\bar{d}} + P_{a\bar{d}}P_{c\bar{b}}. \quad (8.2)$$

Our complex curvature components use the convention

$$R_{a\bar{b}c\bar{d}} = -\mathcal{K}_{a\bar{c}b\bar{d}} + \hat{g}^{p\bar{e}}\mathcal{K}_{ac\bar{e}}\mathcal{K}_{p\bar{b}d}. \quad (8.3)$$

Repeated indices in this formula are summed over all coordinates; repeated horizontal indices below are summed over 1, 2. The inverse coefficients satisfy $\sum_p \hat{g}^{p\bar{e}}\hat{g}_{p\bar{j}} = \delta_{ej}$, so their numerical array is the transpose of the ordinary inverse of the displayed metric matrix. We relate (8.3) to real sectional curvature below, with the fixed convention $g = 2 \operatorname{Re} \hat{g}$.

Proposition 8.1. *In the frame just described, define the horizontal Hermitian form*

$$D = (1 + 2x)h - qhG^{-1}h.$$

The full curvature tensor is

$$R = -\lambda S_I - qS_h - S_V - (S_{D+V} - S_D - S_V) + T_4 + T_3, \quad (8.4)$$

where T_j has exactly j horizontal indices. Its nonzero components, up to the Kähler symmetries and conjugation, are

$$(T_4)_{i\bar{j}k\bar{l}} = -xh_{i\bar{j},k\bar{l}} + x^2G^{p\bar{e}}h_{i\bar{e},k}h_{p\bar{j},\bar{l}}, \quad (8.5)$$

$$(T_3)_{i\bar{j}k\bar{0}} = \sqrt{q}(-h_{i\bar{j},k} + xG^{p\bar{e}}h_{i\bar{e},k}h_{p\bar{j}}). \quad (8.6)$$

Measured by the Euclidean component norms in this frame, uniformly for $\lambda \geq 1$,

$$\|T_4\| \leq C_4 x, \quad \|T_3\| \leq C_3 \sqrt{q}. \quad (8.7)$$

If λ is at least the uniform upper bound for the eigenvalues of h at centered-chart origins, then

$$(1+x)h \leq D \leq (1+2x)h. \quad (8.8)$$

Proof. Write $F(s) = -\log(1 - e^s)$. Direct differentiation gives

$$F' = x, \quad F'' = q, \quad F''' = q(1+2x), \quad F'''' = q(1+6x+6x^2).$$

The centered-chart expression for ψ differs from $-\log(1 - |z|^2)$ by a pluriharmonic function and a constant. Its derivatives of type $(2,1)$ therefore vanish at the center, and its derivatives of type $(2,2)$ are S_I .

For brevity in the derivative tables, put

$$A_{ik\bar{j}} = h_{i\bar{j},k}, \quad B_{ik\bar{j}\bar{l}} = h_{i\bar{j},k\bar{l}}.$$

Before normalizing the vertical vector, the only possibly nonzero third derivatives of type $(2,1)$, up to symmetry, are

$$\mathcal{K}_{ik\bar{j}} = xA_{ik\bar{j}}, \quad \mathcal{K}_{iv\bar{j}} = qh_{i\bar{j}}, \quad \mathcal{K}_{vv\bar{v}} = q(1+2x). \quad (8.9)$$

The fourth derivatives of type $(2,2)$, up to symmetry and conjugation, are

$$\begin{aligned} \mathcal{K}_{ik\bar{j}\bar{l}} &= \lambda(S_I)_{i\bar{j}k\bar{l}} + xB_{ik\bar{j}\bar{l}} + q(S_h)_{i\bar{j}k\bar{l}}, \\ \mathcal{K}_{ik\bar{j}\bar{v}} &= qA_{ik\bar{j}}, \\ \mathcal{K}_{iv\bar{j}\bar{v}} &= q(1+2x)h_{i\bar{j}}, \\ \mathcal{K}_{vv\bar{v}\bar{v}} &= q(1+6x+6x^2). \end{aligned} \quad (8.10)$$

To check that these lists are complete, apply the chain rule to $F(s)$ and use (8.1). In the four-horizontal derivative, the two pairings of mixed second derivatives of s give qS_h ; the third pairing contains $s_{ik}s_{\bar{j}\bar{l}} = 0$. All other terms contain a vanished first or pure second horizontal derivative. In particular, $\mathcal{K}_{ik\bar{v}} = \mathcal{K}_{vv\bar{j}} = 0$, $\mathcal{K}_{ik\bar{v}\bar{v}} = 0$, and the fourth derivatives with one horizontal and three vertical indices vanish. No vanishing of pure third horizontal derivatives is needed.

Substitution into (8.3) gives (8.5) and (8.6). For clarity, all remaining curvature types in the normalized frame are

$$\begin{aligned} R_{i\bar{j}0\bar{0}} &= -(1+2x)h_{i\bar{j}} + q(hG^{-1}h)_{i\bar{j}} = -D_{i\bar{j}}, \\ R_{i\bar{0}k\bar{0}} &= 0, \quad R_{i\bar{0}0\bar{0}} = 0, \quad R_{0\bar{0}0\bar{0}} = -2. \end{aligned} \quad (8.11)$$

The last identity follows because its unnormalized value is

$$-q(1+6x+6x^2) + q(1+2x)^2 = -2q^2,$$

and the four vertical normalization factors multiply to q^{-2} . The tensors S_V and $S_{D+V} - S_D - S_V$ give precisely the vertical and balanced two-horizontal blocks in (8.11). The Kähler symmetries and conjugation now account for every component, proving (8.4).

The lower bound $h \geq cI$ implies

$$\|G^{-1}\| \leq \frac{1}{\lambda + cx}, \quad x^2\|G^{-1}\| \leq \frac{x}{c}.$$

The assumed component-derivative bounds for h , applied to (8.5) and (8.6), prove (8.7) with constants independent of $\lambda \geq 1$ and $x > 0$.

Finally, $G = \lambda I + xh$ commutes with h at the center. On an eigendirection of h with eigenvalue d , the ratio of D to h is

$$1 + 2x - \frac{x(1+x)d}{\lambda + xd}.$$

The subtracted fraction is nonnegative, and it is at most x when $\lambda \geq d$. This proves (8.8). $\square$

Although the logarithmic fiber coordinate was used only for $w \neq 0$, the tensor at the zero section can be checked directly in regular coordinates. A holomorphic fiber change $W = e^{P(z)+a}w$, with constant a and a polynomial of degree at most two, makes the new weight have value zero and vanishing first and pure second derivatives at the base point. Its mixed Hessian remains h . Through total degree four, the terms relevant to the curvature are

$$\lambda\psi + |W|^2 \left(1 + \sum_{i,j} h_{i\bar{j}} z_i \bar{z}_j \right) + \frac{1}{2}|W|^4.$$

At the origin the metric is $\text{diag}(\lambda I, 1)$, all relevant third derivatives vanish, and direct differentiation gives

$$R_{i\bar{j}k\bar{l}} = -\lambda(S_I)_{i\bar{j}k\bar{l}}, \quad R_{i\bar{j}0\bar{0}} = -h_{i\bar{j}}, \quad R_{0\bar{0}0\bar{0}} = -2, \quad (8.12)$$

with all other types zero. In the original fiber coordinate the normalized vertical vector is $e^{-\phi/2}\partial_w$. Thus the formulas above hold at the zero section as well by setting $x = q = 0$, $G = \lambda I$, $D = h$, and $T_3 = T_4 = 0$, and using this regular normalized vertical vector.

8.2 Real curvature and Gram determinants

Let U_1, U_2 be real linearly independent tangent vectors and let u_1, u_2 be their $(1, 0)$ parts, so $U_j = u_j + \bar{u}_j$. For a Hermitian form P , linear in its first argument, define

$$\begin{aligned} a_P &= P(u_1, u_1)P(u_2, u_2) - (\text{Re } P(u_1, u_2))^2, \\ b_P &= \text{Im } P(u_1, u_2), \quad m_P = a_{P+V} - a_P - a_V \quad \text{when } P \text{ is horizontal.} \end{aligned} \quad (8.13)$$

Thus a_P is the determinant of the real Gram matrix for $\text{Re } P$, b_P is the imaginary Hermitian pairing, and m_P is the cross term in that determinant after adding V . For the metric form $\hat{g}$, the real squared area is $4a_{\hat{g}}$ because $g = 2\text{Re } \hat{g}$. For a tensor T with the Kähler symmetries, also put

$$\mathcal{H}_T = T_{u_1 \bar{u}_1 u_2 \bar{u}_2} - \text{Re } T_{u_1 \bar{u}_2 u_1 \bar{u}_2}.$$

Here contraction with complex vectors has its usual multilinear meaning. These definitions allow the horizontal or vertical projection of the real plane to be degenerate.

Lemma 8.2. *For the metric convention $g = 2\text{Re } \hat{g}$ and the curvature convention (8.3),*

$$K_g(\text{span}_{\mathbb{R}}\{U_1, U_2\}) = \frac{\mathcal{H}_R}{2a_{\hat{g}}}. \quad (8.14)$$

Define the matrix

$$\Xi^{a\bar{b}} = u_1^a \bar{u}_2^{\bar{b}} - u_2^a \bar{u}_1^{\bar{b}}. \quad (8.15)$$

For every tensor T with the Kähler symmetries and Hermitian conjugation symmetry,

$$\mathcal{H}_T = -\frac{1}{2}T_{\bar{a}\bar{b}c\bar{d}}\Xi^{ab}\Xi^{c\bar{d}}. \quad (8.16)$$

For every Hermitian form P ,

$$\mathcal{H}_{S_P} = a_P + 3b_P^2. \quad (8.17)$$

Proof. The real squared area of the pair is $4a_{\hat{g}}$, because $g(U_i, U_j) = 2\operatorname{Re}\hat{g}(u_i, u_j)$. In particular $a_{\hat{g}} > 0$ for a real linearly independent pair. Write

$$A = R_{u_1\bar{u}_1u_2\bar{u}_2}, \quad B = R_{u_1\bar{u}_2u_1\bar{u}_2}.$$

The Kähler symmetries in the expansion on $U_j = u_j + \bar{u}_j$ give the real numerator

$$R^{\mathbb{R}}(U_1, U_2, U_2, U_1) = A - B - \bar{B} + A = 2\mathcal{H}_R.$$

Division by the real squared area proves (8.14). Expanding the right-hand side of (8.16) gives the same expression $A - \operatorname{Re} B$ for T . Finally, if $c_P = P(u_1, u_2)$, direct contraction of (8.2) gives

$$\mathcal{H}_{S_P} = P(u_1, u_1)P(u_2, u_2) + |c_P|^2 - 2\operatorname{Re}(c_P^2) = a_P + 3b_P^2.$$

□

For example, the disk potential $-\log(1 - |w|^2)$ has normalized complex curvature component -2 by (8.11). Taking $u_2 = iu_1$ in (8.14) gives real sectional curvature -2 . This fixes the normalization also on complex lines.

Lemma 8.3. *For a positive semidefinite Hermitian form P , $a_P \geq b_P^2$. For the vertical form V one has $a_V = b_V^2$. If P is horizontal and positive semidefinite, m_P is nonnegative, linear and monotone in P , and*

$$m_P \geq 2\sqrt{a_P a_V}. \quad (8.18)$$

When $a_P a_V > 0$, equality in (8.18) holds exactly when the two real Gram matrices of the pair for P and V are proportional.

Proof. Complex Cauchy–Schwarz gives $P(u_1, u_1)P(u_2, u_2) \geq |P(u_1, u_2)|^2$, proving $a_P \geq b_P^2$. Equality holds for V because its complex rank is one. Let A and B be the two-by-two real Gram matrices with entries $\operatorname{Re} P(u_i, u_j)$ and $\operatorname{Re} V(u_i, u_j)$, respectively. Then

$$a_P = \det A, \quad a_V = \det B, \quad m_P = \operatorname{tr}(\operatorname{adj}(A)B).$$

For positive definite A , put $C = A^{-1/2}BA^{-1/2}$. The last expression equals $(\det A)\operatorname{tr} C$, whereas $\sqrt{a_P a_V} = (\det A)\sqrt{\det C}$. The inequality $\operatorname{tr} C \geq 2\sqrt{\det C}$ proves (8.18), and continuity gives the semidefinite case. If both determinants are positive, equality means that C is a scalar matrix, which is exactly the claimed proportionality. The displayed formula for m_P is linear in the entries of P . Its nonnegativity for positive semidefinite P then also proves monotonicity. □

8.3 Error terms and the curvature numerator

Decompose $u_j = (u_{j,H}, u_{j,0})$ in the pointwise frame. Write Ξ_{HH} for the horizontal two-by-two block of Ξ and Ξ_{H0} for its horizontal-to-vertical column. The other mixed block is the negative conjugate transpose of this column. Write Ξ_{mix} for the matrix consisting of both mixed blocks, with all other entries zero. All norms of these blocks below are Euclidean Frobenius norms.

Direct expansion of (8.15) gives

$$\begin{aligned}\|\Xi_{HH}\|^2 &= 2(a_I + b_I^2) \leq 4a_I, \\ \|\Xi_{H\bar{0}}\|^2 &= m_I + 2b_I b_V, \quad \|\Xi_{\text{mix}}\|^2 = 2(m_I + 2b_I b_V) \leq 4m_I.\end{aligned}\tag{8.19}$$

For the second identity, explicitly expand $\|u_{1,H}\overline{u_{2,0}} - u_{2,H}\overline{u_{1,0}}\|^2$. The last inequality follows from $|b_I b_V| \leq \sqrt{a_I a_V} \leq m_I/2$, by Lemma 8.3.

Lemma 8.4. *For every real linearly independent tangent pair,*

$$|\mathcal{H}_{T_4}| \leq Cx a_I, \quad |\mathcal{H}_{T_3}| \leq C\sqrt{q} \sqrt{a_I m_I}.\tag{8.20}$$

More explicitly, the three-horizontal error satisfies

$$|\mathcal{H}_{T_3}| \leq C\sqrt{q} \|\Xi_{HH}\| \|\Xi_{\text{mix}}\|.\tag{8.21}$$

Proof. In (8.16), the tensor T_4 pairs two horizontal blocks of Ξ . The tensor T_3 pairs one horizontal block and one mixed block, since it has exactly three horizontal indices. Cauchy–Schwarz and (8.7) give (8.21) and $|\mathcal{H}_{T_4}| \leq Cx \|\Xi_{HH}\|^2$. Now use (8.19). These estimates remain valid when one of the projected real areas vanishes. $\square$

We can now express the entire real curvature numerator. By (8.17) and $a_V = b_V^2$,

$$\mathcal{H}_{S_V} = 4a_V, \quad \mathcal{H}_{S_{D+V}-S_D-S_V} = m_D + 6b_D b_V.$$

Consequently (8.4) gives

$$\begin{aligned}-\mathcal{H}_R &= \lambda(a_I + 3b_I^2) + q(a_h + 3b_h^2) + 4a_V + m_D + 6b_D b_V \\ &\quad - \mathcal{H}_{T_4} - \mathcal{H}_{T_3}.\end{aligned}\tag{8.22}$$

Its denominator in (8.14) is positive for every real two-plane. It remains to prove that (8.22) is uniformly positive relative to that denominator for sufficiently large fixed λ , and that the quotient is also uniformly bounded above. The mixed term $6b_D b_V$ can have either sign; Section 9 treats it together with the possible degeneration of the projected planes.

9 Uniform pinching on real two-planes

We now prove the curvature bounds in Proposition 2.2. The identity of Section 8 reduces the problem to finite-dimensional algebra. The limiting main expression is nonnegative, but it can vanish on certain real planes. We identify those planes and show that the error with three horizontal indices vanishes to the additional order needed there.

Throughout this section, horizontal forms are extended by zero on the vertical factor of $\mathbb{C}^2 \oplus \mathbb{C}$. We retain $I, V, S_P, a_P, b_P, m_P, \Xi$ from Section 8. Unless otherwise indicated, constants depend only on the fixed bounds $cI \leq h \leq CI$ and the component bounds for the error tensors.

9.1 The mixed real-plane algebra

The leading expression below can vanish even when both projections of the real plane are nonzero. For an explicit algebraic model, take $h = I$, $E = 2I$, and $u_1 = (e_1, 1)$, $u_2 = (ie_1, -i)$, where $e_1 = (1, 0) \in \mathbb{C}^2$. Direct substitution gives

$$a_h + 3b_h^2 + 4a_V + m_E + 6b_E b_V = 4 + 4 + 4 - 12 = 0, \quad \Xi_{\text{mix}} = 0.$$

The horizontal and vertical projections have opposite complex orientations. The next lemma proves nonnegativity for all planes in the stated range of E and shows that every zero has this mixed-block vanishing.

Lemma 9.1. *Let h be a positive definite horizontal Hermitian form, let E be another horizontal Hermitian form with $E \geq h$, and let u_1, u_2 be real-linearly independent vectors in $\mathbb{C}^2 \oplus \mathbb{C}$. Define*

$$P_h(E) = a_h + 3b_h^2 + 4a_V + m_E + 6b_E b_V.$$

If $a_h a_V = 0$, then $P_h(E) > 0$. If $a_h a_V > 0$, put $\tau = |b_h|/\sqrt{a_h}$. For every real β such that $E \leq (2 - \beta)h$,

$$P_h(E) \geq \left(\sqrt{a_h + 3b_h^2} - 2\sqrt{a_V} \right)^2 + 2\beta(1 + \tau)\sqrt{a_h a_V}. \quad (9.1)$$

In particular, $h \leq E \leq 2h$ implies $P_h(E) \geq 0$. If equality holds in this last conclusion, then

$$a_h = a_V > 0, \quad |b_h| = \sqrt{a_h}, \quad b_E b_V < 0,$$

and both mixed blocks $\Xi^{i\bar{0}}$ and $\Xi^{0\bar{i}}$ vanish.

Proof. If $a_V = 0$, then $b_V = 0$, because the vertical factor has complex dimension one. If $a_h = 0$, the two horizontal projections are linearly dependent over $\mathbb{R}$, since h is positive definite. Their pairing under every Hermitian form is therefore real, so $b_E = 0$. Thus $a_h a_V = 0$ implies $b_E b_V = 0$, and

$$P_h(E) \geq a_h + 4a_V + m_h \geq a_{h+V} > 0.$$

Here the last inequality uses the positive definiteness of $h+V$ and the real linear independence of the pair.

Suppose now that $A = a_h > 0$ and $U = a_V > 0$. The Hermitian Gram determinant gives $0 \leq \tau \leq 1$. Make a real change of basis in the plane so that its horizontal projections, denoted u_H, v_H , are real-orthonormal for h . Both ratios $|b_h|/\sqrt{a_h}$ and $|b_E|/\sqrt{a_h}$ are unchanged by this operation. The squared h -norms of $u_H + iv_H$ and $u_H - iv_H$ are $2(1 + \tau)$ and $2(1 - \tau)$, in some order. The absolute difference of their squared E -norms is $4|b_E|$ in the new basis. Comparing one norm from above and the other from below gives

$$\frac{|b_E|}{\sqrt{A}} \leq \frac{(2 - \beta)(1 + \tau) - (1 - \tau)}{2} = \frac{1 + 3\tau}{2} - \frac{\beta(1 + \tau)}{2} \leq \sqrt{1 + 3\tau^2} - \frac{\beta(1 + \tau)}{2}. \quad (9.2)$$

For either order of the two norms, the displayed right side is an upper bound, since $2 - \beta \geq 1$. The last inequality follows from

$$1 + 3\tau^2 - \left(\frac{1 + 3\tau}{2} \right)^2 = \frac{3}{4}(1 - \tau)^2;$$

it is strict unless $\tau = 1$. This argument does not require $\beta \geq 0$.

The mixed Gram inequality (8.18) and the Hermitian Gram inequality yield

$$m_E \geq 2\sqrt{a_E U} \geq 2|b_E|\sqrt{U}.$$

Since $U = b_V^2$, it follows that

$$\begin{aligned} P_h(E) &\geq A + 3b_h^2 + 4U - 4|b_E|\sqrt{U} \\ &\geq \left(\sqrt{A + 3b_h^2} - 2\sqrt{U} \right)^2 + 2\beta(1 + \tau)\sqrt{AU}, \end{aligned}$$

which proves (9.1).

It remains to examine equality when $h \leq E \leq 2h$. The zero-area case already treated shows that $A, U > 0$. Set $s = \sqrt{A + 3b_h^2}$ and $B = |b_E|$. The inequalities above, with $\beta = 0$, give

$$P_h(E) \geq (s - 2\sqrt{U})^2 + 4\sqrt{U}(s - B) \geq 0.$$

Equality forces $s = 2\sqrt{U}$, $B = s$, and equality in the bounds used to reach this expression. The strictness in (9.2) gives $\tau = 1$. Consequently

$$U = A, \quad B = 2\sqrt{A}, \quad b_E b_V < 0, \quad m_E = 2B\sqrt{U}.$$

Since $a_E \geq B^2$ and $m_E \geq 2\sqrt{a_E U}$, these identities force $a_E = B^2$ and equality in the mixed Gram inequality. The horizontal real E -Gram matrix and the vertical real Gram matrix are therefore proportional.

In the real h -orthonormalized basis used above, $\tau = 1$ says that the horizontal projections have the form

$$u_H, \quad \sigma i u_H, \quad \sigma \in \{1, -1\}.$$

Their real E -Gram matrix is a positive scalar multiple of the identity. Proportionality implies that the vertical projections have equal positive norms and real inner product zero. The sign $b_E b_V < 0$ selects the opposite complex orientation, so these projections have the form

$$y, \quad -\sigma i y.$$

Direct substitution now gives

$$\Xi^{i\bar{0}} = u_H^i \overline{-\sigma i y} - (\sigma i u_H^i) \bar{y} = 0.$$

The other mixed block vanishes as well. Under any real change of basis in the plane, Ξ is multiplied by the determinant of that change. The vanishing therefore holds in the original basis. $\square$

9.2 Uniform negativity for large λ

The next statement allows every error tensor with the specified bounds and component types. This uniformity will also give the compactness argument needed after λ has been fixed.

Lemma 9.2. *Fix $0 < c \leq C < \infty$ and $0 \leq C_e < \infty$. There is a finite $\lambda_0 = \lambda_0(c, C, C_e) > 0$ with the following property. Let $\lambda \geq \lambda_0$, $x \geq 0$, and let h be a horizontal Hermitian form with $cI \leq h \leq CI$. Put*

$$q = x(1 + x), \quad G = \lambda I + xh, \quad D = (1 + 2x)h - qhG^{-1}h.$$

Let T_4, T_3 be tensors with the Kähler symmetries and reality condition, supported respectively on components with four and with three horizontal indices, and satisfying

$$\|T_4\| \leq C_e x, \quad \|T_3\| \leq C_e \sqrt{q}.$$

Define

$$R = -\lambda S_I - qS_h - S_V - (S_{D+V} - S_D - S_V) + T_4 + T_3.$$

Then $\mathcal{H}_R < 0$ on every real two-plane.

Proof. Increase the prospective threshold so that $\lambda \geq \max(1, C)$. Then (8.8) gives

$$(1+x)h \leq D \leq (1+2x)h.$$

If the lemma were false, there would be a sequence $\lambda \rightarrow \infty$ of permitted data and real two-planes for which $F = -\mathcal{H}_R \leq 0$. We suppress the sequence index. By (8.22),

$$F = \lambda(a_I + 3b_I^2) + q(a_h + 3b_h^2) + 4a_V + m_D + 6b_D b_V - \mathcal{H}_{T_4} - \mathcal{H}_{T_3}. \quad (9.3)$$

The contracted estimates (8.20) apply to the allowed error tensors by their component types and symmetries.

The regime $q/\lambda \rightarrow 0$. Suppose first that a subsequence has this property. Monotonicity and linearity of the mixed Gram term give $m_D \geq c(1+x)m_I$, while

$$|b_D| \leq \sqrt{a_D} \leq C'(1+x)\sqrt{a_I}.$$

Discarding nonnegative terms in (9.3), we get

$$F \geq \lambda a_I + 4a_V + c(1+x)m_I - C'((1+x)\sqrt{a_I a_V} + x a_I + \sqrt{q} \sqrt{a_I m_I}).$$

Absorb the first product into $2a_V$, and the last into $\frac{c}{2}(1+x)m_I$. Since $q/(1+x) = x$, this gives

$$F \geq [\lambda - C'''((1+x)^2 + x)]a_I + 2a_V + \frac{c}{2}(1+x)m_I.$$

Here $(1+x)^2 + x = O(1+q) = o(\lambda)$, because $\lambda \rightarrow \infty$. Every coefficient is eventually positive, whereas $a_I + a_V + m_I = a_{I+V} > 0$. This contradicts $F \leq 0$.

The regime with λ/q bounded. If the preceding regime has no subsequence, we may pass to one on which q/λ is bounded below by a positive constant. Then λ/q is bounded and $x \rightarrow \infty$. The term qS_h has four horizontal slots, whereas the vertical term S_V already has coefficient one. To balance these two blocks, multiply horizontal vector components by $q^{1/4}$. In the resulting plane choose a real-orthonormal pair for $\text{Re}(I+V)$. The inverse transformation and a real basis change give a basis of the original plane; the latter change multiplies F by a positive factor. Thus its sign is unchanged. The rescaled pairs belong to a fixed compact Stiefel manifold, and their quantities satisfy $a_{I+V} = 1$.

In the rescaled variables, the leading part of (9.3) is

$$L_x = \frac{\lambda}{q}(a_I + 3b_I^2) + P_h(E), \quad E = \frac{D}{\sqrt{q}}. \quad (9.4)$$

The transformed tensors, denoted $\tilde{T}_4, \tilde{T}_3$, satisfy

$$\|\tilde{T}_4\| \leq C' \frac{x}{q} = O(x^{-1}), \quad \|\tilde{T}_3\| \leq C' q^{-1/4} = O(x^{-1/2}). \quad (9.5)$$

Indeed, each horizontal slot supplies a factor $q^{-1/4}$.

The lower bound on D gives $E \geq h$. For an eigenvalue $d \in [c, C]$ of h , the corresponding ratio is

$$\frac{E}{h} = \frac{1+2x}{\sqrt{q}} - \frac{\sqrt{q} d}{\lambda + x d}.$$

Since

$$\frac{1+2x}{\sqrt{q}} = \sqrt{4 + \frac{1}{q}} \leq 2 + \frac{C_1}{x^2}, \quad \frac{\sqrt{q} d}{\lambda + x d} \geq c_1 \frac{x}{\lambda + x},$$

we have the uniform bound

$$h \leq E \leq \left(2 + \frac{C_1}{x^2} - c_1 \frac{x}{\lambda + x}\right) h. \quad (9.6)$$

The constants $c_1 > 0, C_1 < \infty$ depend only on c, C .

Pass to a subsequence on which $h, E, \lambda/q$, and the normalized pair converge. The limiting forms satisfy $h \leq E \leq 2h$. The error contributions tend to zero by (9.5). Lemma 9.1 shows that the two limiting summands in (9.4) are nonnegative. Since $F \leq 0$, both must vanish. In particular $P_h(E) = 0$ at the limit, so $a_h, a_V > 0$ there and the mixed blocks of the limiting Ξ vanish. The comparison

$$c^2 a_I \leq a_h \leq C^2 a_I$$

follows from the corresponding real Gram-matrix bounds. Hence a_I, a_h, a_V are bounded below by positive constants along this subsequence.

We now improve the error estimate on these particular planes. By the contraction formula (8.16), a tensor with exactly three horizontal indices can pair only one horizontal-horizontal block with one mixed block. Thus

$$|\mathcal{H}_{\tilde{T}_3}| \leq C' q^{-1/4} \|\Xi_{HH}\| \|\Xi_{\text{mix}}\| = o(x^{-1/2}). \quad (9.7)$$

Here Ξ_{HH} denotes the block $(\Xi^{i\bar{j}})$, and Ξ_{mix} comprises the blocks $(\Xi^{i\bar{0}})$ and $(\Xi^{0\bar{i}})$. The first block is bounded because the normalized pairs form a compact set, and the mixed blocks tend to zero by the equality case just proved. The explicit tensor factor $q^{-1/4}$ is what makes this an $o(x^{-1/2})$ estimate for the evaluated contribution. No rate of convergence of the planes is being assumed.

Apply (9.1) before taking the limit, with

$$\beta = c_1 \frac{x}{\lambda + x} - \frac{C_1}{x^2}.$$

This choice is permitted by (9.6), including when $\beta < 0$. The positive lower bounds for a_I, a_h, a_V , and boundedness of the normalized pairs, imply

$$L_x \geq c_2 \left(\frac{\lambda}{q} + \frac{x}{\lambda + x} \right) - \frac{C_2}{x^2} \quad (9.8)$$

along the subsequence, for some $c_2 > 0, C_2 < \infty$. For $x \geq 1$, if $\lambda \geq x^{3/2}$, then

$$\frac{\lambda}{q} \geq \frac{x^{3/2}}{2x^2} = \frac{1}{2\sqrt{x}}.$$

If instead $\lambda \leq x^{3/2}$, then

$$\frac{x}{\lambda + x} \geq \frac{1}{\sqrt{x} + 1} \geq \frac{1}{2\sqrt{x}}.$$

Thus (9.8) is bounded below by a positive constant times $x^{-1/2}$ for large x . The evaluated $\tilde{T}_4$ contribution is $O(x^{-1}) = o(x^{-1/2})$, and (9.7) gives the same little-o comparison for $\tilde{T}_3$. It follows that $F > 0$ eventually, a contradiction.

The two regimes exhaust the possible failure sequences. This proves the asserted finite threshold λ_0 , uniformly over all permitted matrices, error tensors, and real planes. $\square$

9.3 Both curvature bounds at a fixed parameter

Proof of Proposition 2.2. The geometric conclusions concerning the domain and completeness were proved in Lemma 7.1. It remains to obtain both uniform real sectional-curvature bounds. By Proposition 8.1, the errors of the actual metric have the types and bounds in Lemma 9.2, with C_e depending only on the chart bounds of Proposition 2.2. Choose any finite $\lambda \geq \lambda_0$. Then $-\mathcal{H}_R > 0$ for every point and every real two-plane. We show that the sectional-curvature quotient is bounded away from zero and from negative infinity.

First restrict x to a bounded interval $[0, X]$. Normalize the pairs for $\text{Re}(I + V)$. The data consisting of x , the matrix h , the component arrays of T_4, T_3 , and the normalized pair range over a closed bounded subset of a finite-dimensional space: impose $cI \leq h \leq CI$, the stated component types and symmetries, and the closed norm bounds $C_e x, C_e \sqrt{q}$. This set is compact. The formulas for G, D, R are continuous on it. Since Lemma 9.2 applies to this entire set, $-\mathcal{H}_R$ has a strictly positive minimum and a finite maximum. The metric $G + V$ is uniformly positive definite and bounded there, so a_{G+V} is bounded above and away from zero. Formula (8.14) therefore gives both a strictly negative upper curvature bound and a finite lower curvature bound on $0 \leq x \leq X$.

To cover the end $x \rightarrow \infty$, use the horizontal rescaling from the preceding proof. Here λ remains fixed: the mixed form $D/\sqrt{q}$ will converge to h , rather than to an arbitrary form between h and $2h$ as allowed in the preceding contradiction argument. The rescaled Hermitian metric is

$$\tilde{g}_x = G/\sqrt{q} + V.$$

For the fixed finite λ ,

$$\frac{G}{\sqrt{q}} - h = \frac{\lambda}{\sqrt{q}}I + \left(\frac{x}{\sqrt{q}} - 1\right)h = O_\lambda(x^{-1}) \quad (9.9)$$

uniformly in the permitted data. Because G commutes with h , the formula for D also gives

$$D = xh + \lambda(1+x)hG^{-1}.$$

Consequently

$$\frac{D}{\sqrt{q}} - h = \left(\frac{x}{\sqrt{q}} - 1\right)h + \frac{\lambda(1+x)}{\sqrt{q}}hG^{-1} = O_\lambda(x^{-1}). \quad (9.10)$$

The rescaled tensor identity now implies

$$\tilde{R}_x = -S_{h+V} + O_\lambda(x^{-1/2}) \quad (9.11)$$

in component norm, uniformly over all allowed data. Indeed, the term λS_I is divided by q , the term qS_h becomes exactly S_h , the mixed term converges by (9.10), and the error tensors satisfy (9.5).

For the limiting metric $h + V$ and tensor $-S_{h+V}$,

$$K = -\frac{1}{2} \left(1 + 3 \frac{b_{h+V}^2}{a_{h+V}}\right) \in [-2, -1/2].$$

For an $(I + V)$ -real-orthonormal pair, $a_{h+V} \geq \min(c, 1)^2$, so these area denominators are uniformly bounded away from zero. The forms $h + V$ also range over a compact bounded family. The metric and tensor estimates (9.9)–(9.11) therefore imply uniform convergence of the corresponding sectional-curvature quotients over all real planes. For sufficiently large $X_\lambda < \infty$, they give, for example,

$$-3 \leq K_g \leq -1/4 \quad (x \geq X_\lambda).$$

Combining this with the compact-range bounds at $X = X_\lambda$ produces finite constants $0 < A \leq B$ for which

$$-B \leq K_g(\sigma) \leq -A < 0$$

for every point and every real two-plane. The tensor identity at $x = 0$, given in (8.12), is computed in a smooth normalized fiber frame; equivalently the same bounds extend to the zero section by smoothness of the metric. This completes Proposition 2.2. $\square$

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